Forcing the Π^1_n-Uniformization Property

Stefan Hoffelner*

November 9, 2026

Abstract

We generically construct a model in which the Π^1_3-uniformization property is true, thus lowering the best known consistency strength from the existence of $M_1^{\#}$ to just ZFC. The forcing construction also works over canonical inner models with Woodin cardinals, which yields, for the first time, universes where the Π^1_{2n}-uniformization property holds for $n > 1$, thus producing models which contradict the natural PD-induced pattern.

1 Introduction

The question of finding nicely definable choice functions for a definable family of sets is an old and well-studied subject in descriptive set theory; see, for instance, [11, 16]. The uniformization problem, first mentioned by N. Lusin in 1930 (see [13]), asks to find choice functions which lie at the same projective level as the set they aim to uniformize. Recall that for $A \subseteq 2^\omega \times 2^\omega$, a *uniformization* of A is a function

$$f : \mathrm{pr}_1(A) \longrightarrow 2^\omega$$

such that

$$\mathrm{graph}(f) \subseteq A.$$

Equivalently, for every $x \in \mathrm{pr}_1(A)$ one has $(x, f(x)) \in A$.

Definition 1.1. *We say that a pointclass Γ has the uniformization property iff every element of Γ admits a uniformization in Γ.*

*The author's research was funded in part by the Austrian Science Fund (FWF) Grant-DOI 10.55776/P37228. This research was also funded in part by the Deutsche Forschungsgemeinschaft (DFG German Research Foundation) under Germany's Excellence Strategy EXC 2044 390685587, Mathematics Münster: Dynamics–Geometry–Structure. For the purpose of open access, the author has applied a CC BY public copyright license to any Author Accepted Manuscript version arising from this submission.

It is a classical result due to M. Kondo [12] that lightface Π^1_1-sets do have the uniformization property, this also yields the uniformization property for Σ^1_2-sets. This is as much as ZFC can prove about uniformization. In the constructible universe L, as shown by J. Addison in [1], Σ^1_n does have the uniformization property for $n \geq 3$, which follows from the existence of a Σ^1_2-good wellorder of the reals; for background on projective wellorders see also [2, 4]. Thus the Π^1_n-uniformization fails for $n \geq 3$. On the other hand, by the celebrated results of Y. Moschovakis (see [15], Theorem 1), $\mathbf{\Delta^1_{2n}}$-projective determinacy implies Π^1_{2n+1}-uniformization, yet the determinacy assumption exceeds in logical strength ZFC. It is known due to W. H. Woodin (see [17]), that $\mathbf{\Delta^1_2}$-projective determinacy implies the existence of $M_1^{\#}$, hence yields an inner model with a Woodin cardinal. As with other regularity properties of the reals like Lebesgue measurability or Baire property, which both follow from PD as well, it is natural to ask whether the Π^1_3-uniformization property bears large cardinal strength as well. We shall answer it negatively.

Theorem. *There is a generic extension of L in which the Π^1_3-uniformization property is true.*

The same uniformization construction also applies over canonical inner models with Woodin cardinals. This can be used to obtain better upper bounds in terms of consistency strength for the Π^1_n uniformization property for odd n. For even $n \geq 4$ we can produce for the first time models where the Π^1_n uniformization property holds true.

Theorem. *Let M_n be the canonical inner model with n Woodin cardinals. Then there is a generic extension of M_n in which the Π^1_{n+3} uniformization property holds true.*

Since the first version of this paper was written, the method developed here has led to further constructions. One combines global Σ-uniformization with a large continuum and a projective well-order [7], and the Σ-side of the local Delfino problem combines related coding methods with projective regularity and uniformization [8]. These later papers refine and apply the forcing method introduced here. The present paper contains the fixed point construction and the hierarchy of allowable iterations needed to force projective Π-uniformization without assuming the corresponding determinacy hypothesis.

Questions concerning the forceability of (local) consequences of PD do have a long tradition in set theory. There is a vast body of literature concerning the forcability of local levels of the projective hierarchy satisfying (Boolean combinations of) the Baire property, the perfect set property or Lebesgue measurability; see for instance [5]. There has been very little progress in the past, however, concerning similar questions for the separation, the reduction and the uniformization property. Indeed, even the question of

whether one can force the Σ^1_3-separation property, which is the weakest of said properties, remained an open problem for 50 years (see [14], Problem 3029).

This article continues this line of research and develops the forcing construction used in the subsequent work mentioned above. It is organized as follows: in the preliminaries section, we briefly introduce the forcings which we will use in the proof and produce a generic extension W of L which will be a well-suited ground model for our needs. We then start to prove the theorems from above. The main idea is to turn the problem of finding a partial order $\mathbb{P}$ which forces the Π^1_3-property into a fixed point problem. We shall define a derivation operator which acts on a specific set of forcing iterations of length ω_1. This operator will be applied transfinitely often, and will produce better and better approximations to the set of forcings we actually want to use in the end. The process is shown to converge in that eventually a fixed point $\mathcal{P}$, i.e. a suitable set of forcings is reached. Forcings which belong to this fixed point $\mathcal{P}$ allow for a certain, seemingly self-referential definition of an ω_1-length iteration which forces the Π^1_3-uniformization property over W. We then follow up, to alter the said process such that it becomes applicable to the canonical inner models M_n with n Woodin cardinals.

There are some similarities to [6], in particular the two proofs rely on a similar ground model W, which is a generic extension of L, and use a similar coding method which relies on a suitably chosen ω_1-sequence of ω_1-Suslin trees. However a more straightforward application of the ideas of [6] will fail badly to produce a model of the Π^1_3-uniformization property. As a consequence, a solution has to necessarily introduce several new ideas in order to succeed. The presentation of those is the goal of this paper.

2 Preliminaries

2.1 Notation

The notation we use will be mostly standard, we hope. We write $\mathbb{P} = (\mathbb{P}_\alpha : \alpha < \gamma)$ for a forcing iteration of length γ with initial segments $\mathbb{P}_\alpha$. The α-th factor of the iteration will be denoted with $\mathbb{P}(\alpha)$. Note here that we drop the dot on $\mathbb{P}(\alpha)$, even though $\mathbb{P}(\alpha)$ is in fact a $\mathbb{P}_\alpha$-name of a partial order.

2.2 The forcings which are used

The forcing notions used in the construction are standard. The preparatory extension is obtained solely by adding branches through a fixed sequence of Suslin trees, and the later coding uses the ground-model version of the ω_1-Cohen forcing together with Jensen–Solovay almost disjoint coding [10].

Recall that a Suslin tree is a normal tree of height ω_1 with countable levels and without an uncountable chain or antichain. Whenever S is a Suslin

tree, we write $\mathbb{S}$ for the forcing whose conditions are the nodes of S, ordered by end-extension. Thus forcing with $\mathbb{S}$ adds a cofinal branch through S.

Definition 2.1. *Let $\vec{S} = \langle S_\xi \mid \xi < \kappa \rangle$ be a sequence of Suslin trees. We call $\vec{S}$* independent *if, for every nonempty finite set $e \subseteq \kappa$, the product*

$$\prod_{\xi \in e} S_\xi$$

is again a Suslin tree.

For $A \subseteq \kappa$ let

$$\mathrm{Br}_A(\vec{S}) = \prod_{\xi \in A}^{\mathrm{fs}} \mathbb{S}_\xi$$

denote the finite-support product of the corresponding branch forcings. When the sequence is fixed, we simply write Br_A.

Lemma 2.2. *Let $\vec{S} = \langle S_\xi \mid \xi < \kappa \rangle$ be an independent sequence of Suslin trees, where $\kappa \leq \omega_1$.*

1. *For every $A \subseteq \kappa$, the forcing Br_A is c.c.c.*
2. *If $A \subseteq B \subseteq \kappa$, then Br_A is a complete subforcing of Br_B, and*

$$\mathrm{Br}_B \cong \mathrm{Br}_A \times \mathrm{Br}_{B \setminus A}\,.$$

3. *If $\xi \notin A$, then*

$$\mathrm{Br}_A \Vdash \text{“}S_\xi \text{ is a Suslin tree.”}$$

Proof. For (1), let $\{p_\alpha \mid \alpha < \omega_1\}$ be a family of conditions in Br_A. After thinning it out, the supports form a Δ-system with finite root r, and the restrictions $p_\alpha \restriction r$ belong to the finite product $\prod_{\xi \in r} S_\xi$. By independence this finite product is Suslin, hence c.c.c., so two of the root restrictions are compatible. Since the remaining parts of their supports are disjoint, the corresponding two conditions in Br_A are compatible.

The factorization in (2) is the usual coordinatewise factorization of a finite-support product. For (3), observe that

$$\mathrm{Br}_A \times \mathbb{S}_\xi \cong \mathrm{Br}_{A \cup \{\xi\}},$$

which is c.c.c. by (1). The standard product characterization of preservation of a Suslin tree now yields the assertion. □

We next recall the almost disjoint coding forcing due to Jensen and Solovay [10]. We identify subsets of ω with their characteristic functions. Let $D = \langle d_\alpha \mid \alpha < \omega_1 \rangle$ be an almost disjoint family of infinite subsets of ω, and let $X \subseteq \omega_1$.

Definition 2.3. *The almost disjoint coding forcing $\mathbb{A}_D(X)$ consists of pairs $(r, R) \in [\omega]^{<\omega} \times [D]^{<\omega}$. We put $(s, S) \leq (r, R)$ if and only if*

1. *$r \subseteq s$ and $R \subseteq S$;*
2. *whenever $\alpha \in X$ and $d_\alpha \in R$, then $s \cap d_\alpha = r \cap d_\alpha$.*

If $G \subseteq \mathbb{A}_D(X)$ is generic and $x_G = \bigcup\{r : (r, R) \in G\}$, then

$$\alpha \in X \quad \Longleftrightarrow \quad x_G \cap d_\alpha \text{ is finite.}$$

Fix a recursive injection $\# : 2^{<\omega} \to \omega$. If $\langle x_\xi : \xi < \omega_1^L \rangle$ enumerates the constructible reals in the canonical order, use $d_\xi = \{\#(x_\xi \restriction k) : k < \omega\}$ and $D = \langle d_\xi : \xi < \omega_1^L \rangle$.

Lemma 2.4. *For every $X \subseteq \omega_1$, the forcing $\mathbb{A}_D(X)$ is Knaster. In particular it is c.c.c. and preserves every ground-model Suslin tree.*

Proof. The Knaster property follows from the usual Δ-system argument applied to the finite side conditions. If T is a Suslin tree, then the product of a Knaster forcing with the c.c.c. forcing $\mathbb{T}$ is c.c.c. Hence $\mathbb{A}_D(X) \times \mathbb{T}$ is c.c.c., which is equivalent to preservation of T as a Suslin tree. □

Finally, we write

$$\mathbb{C}_L(\omega_1) = ({}^{<\omega_1}2)^L$$

for the L-computed ω_1-Cohen forcing, ordered by end-extension. Thus conditions are countable binary sequences belonging to L. Whenever this forcing is used in an outer model of L, the notation refers to this fixed L-poset rather than to a recomputation in the ambient universe.

2.3 The ground model W of the iteration

We now define the ground model over which the main iteration will be carried out. We start with the canonical independent Suslin sequence already present in L and add a cofinal generic branch through every member of that sequence.

The following lemma gives the simultaneous form of Jensen's diamond construction which we use throughout the paper; see also [9]. We use the levelwise product of finitely many trees. This is a dense suborder of the corresponding finite forcing product.

Lemma 2.5. *There is a canonical L-definable sequence*

$$\vec{S} = \langle S_\xi \mid \xi < \omega_1 \rangle$$

of independent normal Suslin trees. The construction is given by a recursion which is uniform over L_{ω_1}.

Proof. Work in L. Fix the canonical L-diamond sequence $\langle a_\alpha \mid \alpha < \omega_1\rangle$ and the canonical L-coding of finite sets of ordinals, countable trees, and subsets of their finite products. We recursively construct coherent countable approximations $\vec{S}^{\,\alpha} = \langle S^\alpha_\xi \mid \xi < \alpha\rangle$ such that each S^α_ξ is a normal splitting tree of height α and $S^\beta_\xi = S^\alpha_\xi \restriction \beta$ whenever $\xi < \beta < \alpha$. At limit stages we take unions.

It remains to describe the passage from α to $\alpha+1$. For old coordinates $\xi < \alpha$, successor stages are handled by putting countably many immediate successors above each node on the top level. Suppose now that α is a nonzero limit. If a_α does not decode as a pair (e, A), where $e \in [\alpha]^{<\omega} \setminus \{\varnothing\}$ and A is a maximal antichain in the product

$$\mathbb{P}^\alpha_e = \bigcup_{\delta<\alpha} \prod_{\xi\in e} (S^\alpha_\xi)_\delta,$$

then we use the $<_L$-least countable family of cofinal branches through each old tree which covers all old nodes, and put one new top node above each of these branches.

If a_α does decode such a pair (e, A), we choose the branch families for the coordinates in e simultaneously so as to seal A. Concretely, by a countable genericity argument in $\prod^{\mathrm{fs}}_{(\xi,n)\in e\times\omega} \mathbb{S}^\alpha_\xi$, choose cofinal branches $b_{\xi,n}$, $\xi \in e$, $n < \omega$, such that they cover all old nodes and such that, for every $\nu : e \to \omega$, the tuple of new top nodes placed above $\langle b_{\xi,\nu(\xi)} \mid \xi \in e\rangle$ extends some member of A. The required dense sets are just the usual ones ensuring cofinality, coverage of old nodes, and the displayed sealing requirement; density of the last kind uses maximality of A in $\mathbb{P}^\alpha_e$. For $\xi \notin e$ we again use the default $<_L$-least covering family. Finally, the new coordinate α is initialized as the canonical finite-support binary tree of height $\alpha+1$. This completes the recursion. Put $S_\xi = \bigcup_{\xi<\alpha<\omega_1} S^\alpha_\xi$. Then each S_ξ is a normal splitting tree of height ω_1 with countable levels.

We show that the sequence is independent. Fix a finite nonempty $e \subseteq \omega_1$ and let

$$\mathbb{P}_e = \bigcup_{\alpha<\omega_1} \prod_{\xi\in e} (S_\xi)_\alpha$$

be the product. This is dense in the usual finite product of the corresponding tree forcings. Let $A \subseteq \mathbb{P}_e$ be a maximal antichain in L. There is a club of $\alpha > \sup(e)$ such that $A \cap (\mathbb{P}_e \restriction \alpha)$ is maximal in $\mathbb{P}_e \restriction \alpha$, and, by coherence of the coding, a club on which the fixed code $x_{e,A}$ for (e, A) restricts to a code for $(e, A \cap (\mathbb{P}_e \restriction \alpha))$. By $\diamondsuit$, choose a limit α in the intersection of these clubs with $a_\alpha = x_{e,A} \cap \alpha$. At stage α the construction seals $A \cap (\mathbb{P}_e \restriction \alpha)$. Hence every node on level α, and therefore every later node, extends a member of this initial antichain. Since A itself is an antichain, $A = A \cap (\mathbb{P}_e \restriction \alpha)$, so A is countable. Thus $\mathbb{P}_e$ has no uncountable antichain. A cofinal branch through $\mathbb{P}_e$ would, by normal splitting, yield an uncountable antichain by choosing

at successor levels nodes incompatible with the branch. Hence $\mathbb{P}_e$ is Suslin. Since e was arbitrary, $\vec{S}$ is independent. The construction is canonical from $<_L$, the diamond sequence, and the fixed coding, and is therefore uniform over L_{ω_1}. □

Lemma 2.6. *Let M be a transitive uncountable model of*

$$\mathsf{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$$

such that

$$\omega_1^M = \omega_1^L = \omega_1.$$

There is a Σ_1-formula

$$\vartheta_{\mathrm{Sus}}(t, \gamma, \eta, \omega_1),$$

independent of M, which defines over M the relation

$$\{(t, \gamma, \eta) : \gamma, \eta < \omega_1 \text{ and } t \in (S_\eta)_\gamma\}.$$

Equivalently, there is a formula $\mathrm{CanSuslin}_L(X, \omega_1)$, which may be taken to be $\Sigma_1(\omega_1)$ over $H(\omega_2)$, such that in every ω_1-preserving outer model U of L,

$$U \models \mathrm{CanSuslin}_L(X, \omega_1) \quad \Longleftrightarrow \quad X = \vec{S}.$$

In particular, the canonical sequence $\vec{S}$ is computed in the same way in all the outer models considered below.

Proof. By Lemma 2.5, the sequence $\vec{S}$ is obtained by a canonical recursion over L_{ω_1}, and the corresponding level relation is $\Sigma_1(\omega_1)$-definable over L_{ω_1}. If M is transitive, satisfies ZFC^- + “$\aleph_1$ exists”, and has $\omega_1^M = \omega_1^L$, then M computes L_{ω_1} correctly. Hence M computes the same canonical diamond sequence, the same coding apparatus, and therefore the same recursive construction of $\vec{S}$. The same Σ_1-formula $\vartheta_{\mathrm{Sus}}(t, \gamma, \eta, \omega_1)$ consequently defines over M the relation $t \in (S_\eta)_\gamma$.

The formulation by $\mathrm{CanSuslin}_L(X, \omega_1)$ is obtained by saying that X is the union of the unique system produced by this canonical recursion. Since every ω_1-preserving outer model of L has the same L_{ω_1}, it satisfies this formula exactly for the true sequence $\vec{S}$. □

Let

$$\mathrm{Br} = \prod_{\xi<\omega_1}^{\mathrm{fs}} \mathbb{S}_\xi$$

be the finite-support branch product, computed in L, and let $H \subseteq \mathrm{Br}$ be L-generic. For every $\xi < \omega_1$ set

$$b_\xi = \bigcup\{p(\xi) : p \in H \text{ and } \xi \in \mathrm{dom}(p)\}.$$

Then b_ξ is a cofinal L-generic branch through S_ξ. In this paper, to *destroy* a Suslin tree always means to add such a cofinal generic branch through it. We define

$$W = L[H].$$

Theorem 2.7. *The model W has the following properties.*

1. Br *is a c.c.c. forcing of size $\aleph_1$. Hence W preserves all cardinals and cofinalities of L, and $W \models \mathsf{CH}$.*
2. *Every S_ξ is destroyed in W, witnessed by the branch b_ξ.*
3. *For $A \subseteq \omega_1$ in L, let $H_A = H \cap \mathrm{Br}_A$. Then H_A is Br_A-generic over L, and*
$$L[H_A] \models \text{“}S_\xi \text{ has a cofinal branch”} \quad \Longleftrightarrow \quad \xi \in A.$$
4. *Every real in W belongs to $L[H_A]$ for some countable $A \subseteq \omega_1$ belonging to L.*

Proof. The first assertion follows from Lemma 2.2. Since $L \models \mathsf{CH}$ and Br has size $\aleph_1$, there are only $\aleph_1$ many nice Br-names for reals, and therefore W still satisfies CH. The definition of H and the dense sets requiring a condition to reach a prescribed level of a prescribed coordinate give (2).

For (3), use the factorization

$$\mathrm{Br} \cong \mathrm{Br}_A \times \mathrm{Br}_{\omega_1 \setminus A}\,.$$

If $\xi \in A$, then $b_\xi \in L[H_A]$. If $\xi \notin A$, then Lemma 2.2 shows that Br_A preserves S_ξ as a Suslin tree, so no cofinal branch through S_ξ belongs to $L[H_A]$.

Finally, let $\dot{x}$ be a nice Br-name for a real. The antichains occurring in $\dot{x}$ are countable, and the union of the finite supports of all conditions appearing in them is therefore a countable set $A \subseteq \omega_1$ in L. Thus $\dot{x}$ is already a Br_A-name, proving (4). □

The relevant inner models of W used below are obtained by retaining selected branch coordinates from H.

Coding forcing. The extensions under consideration preserve ω_1^L, so $\omega_1 = \omega_1^L$. Let $u \in 2^\omega$ and let g be the generic subset of ω_1 added at a new $\mathbb{C}_L(\omega_1)$-coordinate. Let $\varrho_L : ({}^{<\omega_1}2)^L \to \omega_1$ be the canonical bijection. Put

$$h(g) = \{\varrho_L(g \restriction \gamma) : \gamma < \omega_1\},$$
$$I(u,g) = \{\omega\gamma + 2k + u(k) : \gamma \in h(g),\ k < \omega\}.$$

Let $X \subseteq \omega_1$ canonically code $(u, h(g), \langle b_\xi : \xi \in I(u,g)\rangle)$. We describe the localization $Y = Y(X)$ used by $\mathrm{Code}(u,\eta)$, where η specifies the coordinate supplying g.

Use a recursive theory T consisting of ZFC^- with Collection and Choice, existence of ω_1, the relativizations of these axioms to $L[a]$ for each real a, and $\mathrm{HC}^L = L_{\omega_1^L}$. For a transitive $N \models T$ with $u \in N$ and $\omega_1^N = (\omega_1^L)^N = \nu$, put

$$\mathrm{Pat}(N, u, \nu) \quad \Longleftrightarrow \quad N \models \forall \beta < \nu \; \exists \gamma \in [\beta, \nu) \;\; \mathrm{Pattern}(\gamma, u).$$

Here $\mathrm{Pattern}(\gamma, u)$ asserts that for every $k < \omega$ there is a cofinal branch through $S_{\omega\gamma+2k+u(k)}$. The predicate includes the displayed requirements on N, u, ν. The trees are defined by Lemma 2.6.

In $L[X]$, collapse an elementary submodel of a sufficiently large H_χ of size $\aleph_1$ containing X, u, ω_1 and all ordinals below ω_1. Its collapse N satisfies $\mathrm{Pat}(N, u, \omega_1)$. Choose a bijection $e : \omega_1 \to N$ and code membership and satisfaction via e by $Z \subseteq \omega_1$, marking the indices of u and ω_1. Use finite tuple codes ordered by maximum. In $L[Z]$, take the club of closure points ν at which $Z \cap \nu$ codes membership and satisfaction of $e[\nu]$ via $e \restriction \nu$, $e[\nu] \prec N$, with $u, \omega_1 \in e[\nu]$ and $e[\nu] \cap \omega_1 = \nu$. Such a club is obtained by closing under the Skolem maps, e and its inverse. At these points the collapse N_ν satisfies $\mathrm{Pat}(N_\nu, u, \nu)$. Let E consist of the nonzero limit points of this club and 0. If $c \in E$, write $d_c = \min(E \setminus (c+1))$ and let w_c be the $<_L$-least real coding a well-order of type d_c. Set

$$Y(X) = \{2 \cdot \xi + 1 : \xi \in Z\} \cup \{2 \cdot (c + k) : c \in E, \; k < \omega, \; w_c(k) = 1\}.$$

The intervals $[c, c + \omega)$ are disjoint. Make the choices in $L[X]$ and $L[Z]$ by their canonical well-orders, so the same X gives the same Y in all models M with $\omega_1^M = \omega_1$. This is the localization argument of David [3]. The second factor of $\mathrm{Code}(u, \eta)$ is $\mathbb{A}_D(Y(X))$.

Let $\Psi(r, u)$ assert that for every countable transitive $M \models T$ with $r, u \in M$ and $\omega_1^M = (\omega_1^L)^M$, one has

$$M \models \exists N \in L[r] \;\; \mathrm{Pat}(N, u, \omega_1^M).$$

A real coding M can also code its satisfaction relation. The well-foundedness requirement is Π^1_1 and the remaining tests on the code are arithmetic. Hence Ψ is Π^1_2, and $\Phi(u) \iff \exists r \, \Psi(r, u)$ is Σ^1_3.

Lemma 2.8. *The generic real r of the almost disjoint factor of* $\mathrm{Code}(u, \eta)$ *satisfies $\Psi(r, u)$. Conversely, in an extension preserving ω_1, if $\Psi(r, u)$ holds, then for unboundedly many $\gamma < \omega_1$ and every $k < \omega$, $L[r]$ contains a cofinal branch through $S_{\omega\gamma+2k+u(k)}$.*

Proof. Given a countable transitive M as in the definition of Ψ, put $\nu = \omega_1^M$. The definition of D and $\mathrm{HC}^L = L_{\omega_1^L}$ give the correct $D \restriction \nu$ in M. The almost disjoint real decodes $Y \cap \nu$ in $L[r]^M$. If $\nu \notin \mathrm{Lim}(E)$, let $c = \max(E \cap \nu)$. Then $c + \omega < \nu \leq d_c$. The even part gives $w_c \in M$, contrary to $\nu = \omega_1^M$. At

a limit point of E the odd part yields the code of N_ν, so the collapse lies in $L[r]^M$ and is as required.

For the converse, use $F = H_\chi^{L[r,u]}$ for sufficiently large regular χ. This model satisfies T and has $\omega_1^F = (\omega_1^L)^F = \omega_1$. If the conclusion in the definition of Ψ failed at F, a countable elementary submodel of F containing r, u, ω_1 would collapse to a countable transitive model witnessing failure. Thus F contains such an $N \in L[r]$. The trees at ω_1 are computed correctly, and the branches in N belong to $L[r]$. Finally, once r witnesses $\Phi(u)$ it continues to witness it by Shoenfield absoluteness. □

2.4 Coding iterations and their initial segments

For $E \in L$ of size at most $\aleph_1$, put

$$\mathbb{C}_E = \mathrm{Fn}_{<\omega_1}(E \times \omega_1, 2)^L.$$

Use the Cohen index sets $E_i = [\omega_1 \cdot i, \omega_1 \cdot (i+1))$ and put $E_{<\lambda} = \bigcup_{i<\lambda} E_i$. At stage i we first force with $\mathbb{C}_{E_i}$ and then with an almost disjoint factor $\dot{\mathbb{A}}_i$, which may be trivial. Thus $\mathbb{P}_i$ is the initial segment through the first i stages, and $\mathbb{P}_i * \mathbb{C}_{E_i}$ is the forcing just before $\dot{\mathbb{A}}_i$.

Whether to code at stage i, the real u_i to be coded, the index $\eta_i \in E_i$, and the isomorphism $\varphi_i : \mathbb{C}_{E_i} \cong \mathbb{C}_{E_i} \restriction c_i$ are given by $\mathbb{P}_i$-names. Each possible value of φ_i is in L and is induced by bijections on the individual Cohen coordinates, as in the proof below, with at most countably many nonidentity coordinates. If K_i is $\mathbb{C}_{E_i}$-generic, let g_i be the η_i-th Cohen subset of ω_1 obtained from the filter generated by $\varphi_i[K_i]$. At a coding stage put

$$I_i = I(u_i, g_i), \qquad X_i = \mathrm{code}\big(u_i, h(g_i), \langle b_\xi : \xi \in I_i \rangle\big)$$

and use $\mathbb{A}_D(Y(X_i))$, possibly restricted below a condition a_i. The name for a_i is a $\mathbb{P}_i * \mathbb{C}_{E_i}$-name. At other stages we still use $\mathbb{C}_{E_i}$, but $\dot{\mathbb{A}}_i$ is trivial and $I_i = \varnothing$.

For $\lambda \le \omega_1$, write the forcing as

$$\mathbb{P}_\lambda = \mathbb{C}_{E_{<\lambda}} * \dot{\mathbb{R}}_\lambda, \qquad \mathbb{R}_\lambda = \operatorname*{fs}_{i<\lambda} * \dot{\mathbb{A}}_i. \tag{1}$$

Here the names for $\dot{\mathbb{A}}_i$ are translated from $\mathbb{P}_i * \mathbb{C}_{E_i}$. The later Cohen forcings are ground-model partial orders and the names for the earlier factors use only the earlier Cohen coordinates. Moreover the order on $\mathbb{A}_D(Y) \restriction a$ is determined by Y and a. Thus the later Cohen forcings can be moved before the earlier almost disjoint factors. At a limit, (1) uses the countable-support Cohen product computed in L and the finite-support almost disjoint iteration. For $\lambda = \omega_1$, the Cohen product is isomorphic to $\mathrm{Fn}_{<\omega_1}(\omega_1 \times \omega_1, 2)^L$.

Use the dense suborder of $\mathbb{R}_\lambda$ consisting of the finite functions f assigning pairs in $[\omega]^{<\omega} \times [\omega_1]^{<\omega}$ to stages in its support. An omitted stage denotes the

greatest condition of its factor. At a successor one strengthens the preceding condition to decide the name for the next pair. At a limit a finite support is bounded. This proves density by induction. The same induction shows that these dense suborders agree in two models $N_0 \subseteq N_1$ with the same ordinals if the factors have the same conditions and order in common generic extensions by the preceding factors. At a successor, if the two models decided membership or order of a pair differently, this would contradict agreement in a generic extension of the larger model. At a limit each such decision is made in a bounded initial segment.

The complete embeddings of initial segments and the corresponding translations of names are part of the presentation. A tail in this presentation has a fixed countable length and uses Cohen coordinates disjoint from those of the preceding iteration. The bijections of Cohen index sets used to present products and quotients belong to L.

Lemma 2.9. *The form* (1) *is preserved by the following operations.*

1. *For $\beta \leq \lambda$ and $G_\beta \subseteq \mathbb{P}_\beta$ generic,*

$$\mathbb{P}_\beta \lessdot \mathbb{P}_\lambda, \qquad \mathbb{P}_\lambda / G_\beta \simeq \mathbb{C}_{E_{<\lambda} \setminus E_{<\beta}} * \dot{\mathbb{R}}_{[\beta,\lambda)}.$$

   On the dense suborder the projection is

$$\pi_\beta(c, f) = \big(c \restriction (E_{<\beta} \times \omega_1), f \restriction \beta\big).$$

2. *A tail defined below $0 < d \in \mathrm{RO}(\mathbb{P})$ lifts to a $\mathbb{P}$-name with the same Cohen factors and with*

$$\dot{\mathbb{A}}_i^{\uparrow} = \begin{cases} \dot{\mathbb{A}}_i, & \text{on } d, \\ \mathbf{1}, & \text{on } \neg d. \end{cases}$$

   For p in the dense suborder, $\mathbb{P}_\lambda \restriction p$ again has form (1)*, with initial segments $\mathbb{P}_\beta \restriction \pi_\beta(p)$, $\beta \leq \lambda$.*

3. *Tails $\mathbb{Q}, \mathbb{R}$ over the same preceding forcing admit product presentations in either order, on disjoint Cohen index sets. With $\mathbb{R}$ first, $\mathbb{R} \times \mathbb{Q}_i$ is an initial segment, and*

$$\mathbb{R} \times \mathbb{Q}_i \lessdot \mathbb{R} \times \mathbb{Q}.$$

   If $\dot{\mathbb{T}}$ is a $\mathbb{Q}_i$-name for a continuation of fixed countable length, lifted by (2) if necessary, its translation $\dot{\mathbb{T}}_{\mathbb{R}}$ satisfies

$$(\mathbb{R} \times \mathbb{Q}_i) * \dot{\mathbb{T}}_{\mathbb{R}} \simeq \mathbb{R} \times (\mathbb{Q}_i * \dot{\mathbb{T}})$$

   over $\mathbb{R} \times \mathbb{Q}_i$, preserving all previously specified embeddings and translations of names.

Proof. (1) Use

$$\mathbb{C}_{E_{<\lambda}} \cong \mathbb{C}_{E_{<\beta}} \times \mathbb{C}_{E_{<\lambda}\setminus E_{<\beta}}.$$

If $p = (c, f)$ and $q = (c', f') \leq \pi_\beta(p)$, then

$$(c' \cup c \restriction ((E_{<\lambda} \setminus E_{<\beta}) \times \omega_1),\ f' \cup f \restriction [\beta, \lambda)) \leq p, q.$$

After the earlier coordinates are strengthened, the values of f at later stages are still conditions. This proves completeness and the quotient formula, also at limits.

(2) The lift is obtained by mixing names. For the restriction below (c, f), choose in L, for each $\eta \in \operatorname{supp}(c)$,

$$\sup(\operatorname{dom} c_\eta) + \omega < \nu_\eta < \omega_1,$$
$$\tau_\eta : \omega_1 \longrightarrow \omega_1 \setminus \operatorname{dom} c_\eta \quad \text{a bijection}, \qquad \tau_\eta \restriction [\nu_\eta, \omega_1) = \mathrm{id}.$$

Set $\tau_\eta = \mathrm{id}$ elsewhere. Then

$$\psi_c : \mathbb{C}_E \cong \mathbb{C}_E \restriction c, \qquad \psi_c(q)_\eta = c_\eta \cup (q_\eta \circ \tau_\eta^{-1}).$$

Compose the restrictions of ψ_c to the blocks E_i with φ_i and replace $\mathbb{A}_i$ by $\mathbb{A}_i \restriction f(i)$ for $i \in \operatorname{dom}(f)$. Below (c, f) this preserves $g_i, X_i, Y(X_i)$. By (1) the maps commute with all initial-segment projections, and a composition of such maps again changes each coordinate only on a bounded set.

(3) Reindex the Cohen coordinates disjointly in L and translate each tail's names from its own initial segments. In either order the almost disjoint factors have identical conditions and orders, so induction on the dense suborders gives the product isomorphisms. For a continuation of $\mathbb{R} \times \mathbb{Q}_i$, retain both index maps, extend the $\mathbb{Q}_i$-map, and translate the Boolean value in (2) by the fixed embedding of $\mathbb{Q}_i$. This gives the displayed isomorphism over $\mathbb{R} \times \mathbb{Q}_i$. The same induction applies to translations along an isomorphism and its inverse. □

Lemma 2.10. *For $\lambda \leq \omega_1$,*

$$\operatorname{dens}(\mathbb{P}_\lambda) \leq \aleph_1, \qquad \mathbb{P}_\lambda \textit{ preserves cardinals and } \mathsf{CH}.$$

If $G_{\omega_1} \subseteq \mathbb{P}_{\omega_1}$ is generic and $G_\beta = G_{\omega_1} \cap \mathbb{P}_\beta$, then

$$(2^\omega)^{W[G_{\omega_1}]} = \bigcup_{\beta<\omega_1} (2^\omega)^{W[G_\beta]}.$$

Proof. Put $\mathbb{C} = \mathbb{C}_{E_{<\lambda}}$ and let $K \subseteq \mathbb{C}$ be generic over W. Since $L \models |\mathbb{C}| \leq \aleph_1$ and $\mathbb{C}$ is countably closed in L, Easton's lemma applied to Br gives

$$(\mathrm{Ord}^\omega)^{W[K]} = (\mathrm{Ord}^\omega)^W.$$

Thus $\mathbb{C}$ preserves cardinals and CH over W. In $W[K]$, $\mathbb{R}_\lambda$ is c.c.c. with a dense set of finite functions of size $\leq \aleph_1$. Hence $\mathbb{C} * \dot{\mathbb{R}}_\lambda$ has the asserted properties.

For $z \in (2^\omega)^{W[G_{\omega_1}]}$, take a nice $\mathbb{R}_{\omega_1}$-name $\dot{z}$ in the dense suborder. Its ordinal code belongs to W, and

$$\operatorname{supp}(\dot{z}) = \bigcup\{\operatorname{dom}(p) : (\check{n}, p) \in \dot{z}\} \subseteq \beta \quad \text{for some } \beta < \omega_1.$$

The dense suborders of $\mathbb{R}_\beta$ in $W[K \restriction E_{<\beta}]$ and $W[K]$ agree. Thus $\dot{z}$ is already a name over $\mathbb{P}_\beta$, and $z \in W[G_\beta]$. □

Lemma 2.11. *In every extension by an iteration of the form* (1)*,*

$$\{u \in 2^\omega : \Phi(u)\} = \{u_i : i < \lambda,\ i \text{ is a coding stage}\}.$$

Proof. The inclusion from right to left follows from Lemma 2.8, also for factors restricted below conditions. Conversely, let r witness $\Phi(u)$. Work first in $W[K]$, with $K \subseteq \mathbb{C}_{E_{<\lambda}}$ generic, and let $G \subseteq \mathbb{R}_\lambda$ be the remaining generic.

Choose names for r, u and the stage parameters using countable antichains in the dense suborder. Let $D_i \subseteq i$ collect the supports of the parameter names at stage i, including a_i. Put

$$J_0 = \operatorname{supp}(\dot{r}) \cup \operatorname{supp}(\dot{u}),$$
$$J_{n+1} = J_n \cup \bigcup_{i \in J_n} D_i, \qquad J = \bigcup_{n<\omega} J_n \in [\lambda]^{\leq\omega}.$$

The names for r, u and for the parameters at stages in J have a common countable ordinal code z. By Easton's lemma and the c.c.c. of Br,

$$z \in W, \qquad z \in L[H_B] \quad \text{for some } B \in [\omega_1]^{\leq\omega} \cap L.$$

Since $D_i \subseteq J \cap i$ for $i \in J$, induction gives

$$\mathbb{R}_{J\cap i} \lessdot \mathbb{R}_i \quad (i \leq \lambda), \qquad \dot{r}^{G_J} = r, \quad \dot{u}^{G_J} = u, \quad G_J = G \cap \mathbb{R}_J.$$

At successors the factor at a stage in J is given by an $\mathbb{R}_{J\cap i}$-name; at limits use finite supports and the coherent projections.

Fix $\xi \notin B \cup \bigcup_{i\in J} I_i$ and put

$$M_\xi = L[K][H \restriction (\omega_1 \setminus \{\xi\})].$$

Then $z \in M_\xi$. In M_ξ form $\widetilde{\mathbb{R}}^\xi_J$ by following $\mathbb{R}_J$ until the first stage whose names fail to determine unique data or whose I_i contains ξ, and taking trivial factors from that stage onward. The test $\xi \in I_i$ precedes the formation of

X_i, so the factors before that stage have the same dense suborders over M_ξ and $W[K]$. In $W[K]$, induction gives

$$d_\xi = \bigwedge_{i\in J} \|\xi \notin \dot{I}_i\|_{\mathrm{RO}(\mathbb{R}_J)},$$
$$\widetilde{\mathbb{R}}^\xi_J \lessdot \mathbb{R}_J, \qquad \widetilde{\mathbb{R}}^\xi_J \restriction d_\xi \simeq \mathbb{R}_J \restriction d_\xi.$$

At limits take meets. Our choice of ξ makes G_J meet $\mathbb{R}_J \restriction d_\xi$, inducing an M_ξ-generic $\widetilde{G} \subseteq \widetilde{\mathbb{R}}^\xi_J$ with

$$r, u \in M_\xi[\widetilde{G}].$$

The Cohen product preserves the c.c.c. of ground-model branch products, whence

$$L[K] \models \mathrm{Br}_{\omega_1\setminus\{\xi\}} \times \mathbb{S}_\xi \text{ is c.c.c.}$$

All nontrivial factors of $\widetilde{\mathbb{R}}^\xi_J$ are σ-centered. Hence

$$M_\xi[\widetilde{G}] \models \text{“}S_\xi \text{ is Suslin.”}$$

Let $\mathcal{C} = \{i \in J : i \text{ is a coding stage}\}$. Distinct Cohen coordinates differ on an unbounded set, and the coordinate isomorphisms change only bounded initial segments. Thus their common initial segments have bounded length, and

$$\sup(h(g_i) \cap h(g_j)) < \omega_1 \qquad (i \neq j \in \mathcal{C}).$$

Suppose no stage codes u. Choose $\delta < \omega_1$ with

$$B \subseteq \delta, \qquad h(g_i) \cap h(g_j) \subseteq \delta \quad (i \neq j \in \mathcal{C}).$$

By Lemma 2.8, choose $\gamma > \delta$ such that

$$\forall k < \omega \quad L[r] \models \text{“}S_{\omega\gamma+2k+u(k)} \text{ has a cofinal branch.”}$$

There is at most one $i \in \mathcal{C}$ with $\gamma \in h(g_i)$. If such i exists, choose k with $u_i(k) \neq u(k)$; otherwise put $k = 0$. Then

$$\xi = \omega\gamma + 2k + u(k) \notin B \cup \bigcup_{i\in J} I_i.$$

But $L[r] \subseteq M_\xi[\widetilde{G}]$, and S_ξ is Suslin in that model, a contradiction. □

Up to the specified presentations, the bounded iterations, their restrictions below conditions, and the names for their tails form a set. Choose dense suborders carried by subsets of ω_1. Their orders, the specified embeddings, and the names for stage parameters can then be encoded by subsets of fixed sets of conditions and names. For a tail whose countable length is given by a name, first work below a condition deciding the length and apply Lemma 2.9(2). Thus the forcing predicates used below may be indexed by tail names of fixed countable length.

2.5 Motivation for the allowability hierarchy

We call the coding iterations of countable length from Subsection 2.4, possibly restricted below a condition, and their quotients over extensions by initial segments *allowable*. Given the notion of allowable forcing, we now form successive approximations to the class of forcings which we eventually want to use. We first describe the idea leading to these approximations.

Fix a recursive enumeration

$$\langle \varphi_m(v_0, v_1) \mid m \in \omega \rangle$$

of the Π^1_3-formulas with two free real variables, and put

$$A_m := \{(x, y) \in (2^\omega)^2 : \varphi_m(x, y)\}.$$

We write f_m for the uniformizing function which we intend to define for A_m, and write $f(m, x)$ for $f_m(x)$. The goal is to choose, for every $m \in \omega$ and every real x for which

$$A_{m,x} := \{y \in 2^\omega : (x, y) \in A_m\}$$

is nonempty, a real $f(m, x)$ such that

$$(x, f(m, x)) \in A_m,$$

and to code every triple (x, y, m) such that

$$y \neq f(m, x) \quad \text{and} \quad (x, y) \in A_m$$

into $\vec{S}$. The final forcing will have length ω_1, and each of its countable initial segments will be allowable. Since the statement

$$\text{“}(x, y, m) \text{ is coded into } \vec{S}\text{”}$$

is Σ^1_3, the requirement

$$(x, y) \in A_m \quad \text{and} \quad (x, y, m) \text{ is not coded into } \vec{S}$$

is Π^1_3. The bookkeeping of the final iteration will ensure that, whenever $A_{m,x} \neq \varnothing$, exactly one such y remains.

The idea behind the definition can be seen in the following toy example. Work in W, let $x \in W$, and suppose that

$$A_{m,x} = \{y_0, y_1\}.$$

Assume in addition that no allowable forcing over W adds a new member to $A_{m,x}$. We consider only the problem of defining $f(m, x)$ and omit, for the moment, the interaction with the other sections.

Call $y \in A_{m,x}$ *eliminable* if there is an allowable forcing $\mathbb{P}$ such that

$$\mathbb{P} \Vdash (x, y) \notin A_m.$$

Call y *persistent* if it is not eliminable. Thus, if y_0 is persistent, we may reserve y_0 as the value of $f(m, x)$ and code (x, y_1, m) into $\vec{S}$. If y_0 is eliminable, witnessed by an allowable forcing $\mathbb{P}$, we consider y_1. If y_1 is persistent, we reserve y_1 and code (x, y_0, m).

It remains to consider the case in which both values are eliminable. Let $\mathbb{P}$ and $\mathbb{Q}$ witness, respectively, that

$$\mathbb{P} \Vdash (x, y_0) \notin A_m \qquad \text{and} \qquad \mathbb{Q} \Vdash (x, y_1) \notin A_m.$$

By Lemma 2.9, the product

$$\mathbb{P} \times \mathbb{Q}$$

is isomorphic to an allowable forcing. Using this presentation of the product, upward absoluteness gives

$$\mathbb{P} \times \mathbb{Q} \Vdash (x, y_0) \notin A_m \wedge (x, y_1) \notin A_m.$$

By the assumption that no new member is added to the section, the product forces

$$A_{m,x} = \varnothing.$$

In this case there is no value to uniformize at x.

This argument isolates the two operations which are needed later:

reserve a value which is persistent under the forcings under consideration;
combine eliminating forcings by taking products. (2)

The difficulty is that an eliminating forcing may add codes which interfere with other sections. Thus it is not enough to use arbitrary allowable forcings as witnesses. We need a class $\mathcal{P}$ of allowable forcings such that the following rule produces forcings in $\mathcal{P}$ again. For every task (x, m) considered by the bookkeeping, either

$$\exists y \in A_{m,x}\ \forall \mathbb{Q} \in \mathcal{P}\big(\mathbb{Q} \text{ extending the current forcing} \to \mathbb{Q} \Vdash (x, y) \in A_m\big),$$

in which case one such y is reserved and the other values are coded, or every value under consideration can be eliminated by a forcing in $\mathcal{P}$.

This turns the construction into a fixed point problem. Let

$$\mathcal{A}_0^W$$

be the class of allowable iterations. The first refinement searches for values which are persistent under $\mathcal{A}_0^W$-extensions; the resulting iterations form $\mathcal{A}_1^W$.

The next refinement searches also for values which are persistent under $\mathcal{A}_1^W$-extensions, and so on. We obtain a decreasing hierarchy

$$\mathcal{A}_0^W \supseteq \mathcal{A}_1^W \supseteq \cdots \supseteq \mathcal{A}_\alpha^W \supseteq \cdots .$$

In the first refinement we do not code a minimizer if all values of least rank are already present.

The reason for continuing the construction is the usual moving-target problem. In the definition of 1-allowability we test persistence under 0-allowable extensions, but the iteration which we produce is itself 1-allowable. We should therefore also test persistence under 1-allowable extensions. Repeating this argument leads to the transfinite hierarchy defined in the next subsection.

2.6 Persistence and the choice of a uniformizing value

Use the coding iterations of Subsection 2.4, with the specified complete embeddings of their initial segments. We work in extensions M by such an initial segment, sometimes restricted below a condition. An iteration over M is a quotient with the presentation given in that subsection. The phrase *code* (m, x, y) means to code the real $\langle x, y, m\rangle$. Put

$$A(m, x, y) \quad \Longleftrightarrow \quad (x, y) \in A_m.$$

We use two absoluteness facts for A. A witness to $\neg A(b)$ in the extension by one factor of a product remains a witness in the product extension. If $\neg A(b)$ holds in the final extension, its witness gives $\neg A(b)$ in the extension by every sufficiently long proper initial segment of the iteration. Here $b = (m, x, y)$ belongs to the model in which membership is tested. In the construction over L, write $\neg A(b)$ as $\exists z\, \psi(b, z)$ with $\psi \in \Pi^1_2$ and apply Shoenfield absoluteness to ψ. The second assertion also uses the fact that every real belongs to an extension by a bounded initial segment.

Choice functions. For each extension M under consideration we include a name for a choice function J_M on its nonempty sets of reals, so $J_M(B) \in B$. Along each specified inclusion $M \subseteq N$, require $J_N(B) = J_M(B)$ for $B \in \mathrm{dom}(J_M)$. These are names over the given forcings.

Lemma 2.12. *Let $\mathbb{Q}, \mathbb{R}$ be countable iterations over M. For their product extensions put*

$$M_{ij} = M[G_{\mathbb{Q}_i}, G_{\mathbb{R}_j}], \qquad \mathcal{D}_{ij} = \big(\mathcal{P}(\mathbb{R}) \setminus \{\varnothing\}\big)^{M_{ij}}.$$

Coherent choice functions on the extensions M_{i0} and M_{0j}, agreeing on $M = M_{00}$, extend by names to functions

$$J_{ij} : \mathcal{D}_{ij} \longrightarrow \mathbb{R}^{M_{ij}}, \qquad J_{ij}(B) \in B,$$
$$i \le i',\ j \le j' \quad \Longrightarrow \quad J_{i'j'} \restriction \mathcal{D}_{ij} = J_{ij}.$$

If coherent functions are prescribed for $i \le i_0,\ j \le j_0$, the extension can be chosen to agree with them. The assertions remain valid under restrictions below conditions, mixing of names, quotients by $\mathbb{Q}_{i_0} \times \mathbb{R}_{j_0}$, and appending an iteration to either factor of a specified product.

Proof. Order pairs coordinatewise, with meet $a \wedge b$ and join $a \vee b$. We first show

$$\mathcal{D}_a \cap \mathcal{D}_b = \mathcal{D}_{a \wedge b}.$$

For incomparable a, b, factor over $N = M_{a \wedge b}$. Fix a bijection $e : \theta \to \mathbb{R}^N$ in N. The intersection lemma for the mutually generic quotients, applied first to each element of B and then to $e^{-1}[B]$, gives

$$B \in \mathcal{D}_a \cap \mathcal{D}_b \implies B \subseteq \mathbb{R}^N \implies e^{-1}[B] \in N \implies B \in \mathcal{D}_{a \wedge b}.$$

The comparable case is immediate. Consequently

$$\ell(B) = \big(\min\{i : \exists j\ B \in \mathcal{D}_{ij}\},\ \min\{j : \exists i\ B \in \mathcal{D}_{ij}\}\big)$$

satisfies

$$B \in \mathcal{D}_{\ell(B)}, \qquad B \in \mathcal{D}_a \iff \ell(B) \le a.$$

Indeed, take the meet of pairs witnessing the two coordinate minima.

Let J^0_a denote the prescribed functions and let $\lhd_a$ be a named well-order of $\mathbb{R}^{M_a}$. For $B \in \mathcal{D}_a$, set

$$J_a(B) = \begin{cases} J^0_{\ell(B)}(B), & \text{if } \ell(B) \text{ is prescribed,} \\ \min_{\lhd_{\ell(B)}} B, & \text{otherwise.} \end{cases}$$

The set of pairs with prescribed functions is downward closed, so all prescribed values are kept and $a \le b \Rightarrow J_b \restriction \mathcal{D}_a = J_a$. These are names: for $a \le b$, membership in the smaller extension is expressed by

$$\|\dot{B} \in M_a\| = \bigvee_{\dot{C}} \|\dot{B} = \iota_{ab}(\dot{C})\|,$$

where ι_{ab} is the specified translation and $\dot{C}$ ranges over a fixed set of names for sets of reals over $\mathbb{Q}_{a_0} \times \mathbb{R}_{a_1}$, where $a = (a_0, a_1)$. For $B \in \mathcal{D}_b$, the minima defining $\ell(B)$ can be computed using only pairs $a \le b$.

If $B \in \mathcal{D}_c$ is old and occurs at a new pair a, then $B \in \mathcal{D}_{a \wedge c}$, with $a \wedge c$ old. Hence $\ell_{\text{new}}(B) = \ell_{\text{old}}(B)$. For $s = (i_0, j_0)$, restricting to pairs $a \ge s$ instead gives

$$\ell_{\ge s}(B) = \ell(B) \vee s.$$

If $\ell(B) \vee s \neq \ell(B)$, then $\ell(B) \vee s$ has first coordinate i_0 or second coordinate j_0, and the choice is that of the original product; otherwise $\ell(B) \vee s = \ell(B)$, so the original choice applies. This gives compatibility with quotients by $\mathbb{Q}_{i_0} \times \mathbb{R}_{j_0}$ and with both product orders.

For restrictions use the specified embeddings and

$$(\mathbb{Q} \times \mathbb{R}) \restriction (q, r) = (\mathbb{Q} \restriction q) \times (\mathbb{R} \restriction r).$$

Below d retain the prescribed names; on $\neg d$ use a coherent extension and mix the names. Finally, for a continuation $\dot{\mathbb{T}}$ over $\mathbb{Q}_{i_0}$ given below d, Lemma 2.9(2) supplies $\dot{\mathbb{T}}^{\uparrow}$ with trivial almost disjoint factors off d. Apply the construction to $\mathbb{Q}_{i_0} * \dot{\mathbb{T}}^{\uparrow}$ and the other factor. Since $\ell(B)$ does not change, the earlier choices are kept, and Lemma 2.9(3) preserves the specified embeddings. $\square$

We work with the set of all countable iterations under consideration, together with their specified embeddings and the names for their choice functions. The same forcing may occur with different data. For each presentation $\mathbb{H}$, fix a dense suborder $D_{\mathbb{H}}$ of size at most $\aleph_1$. Names for reals may be taken in the form

$$\dot{r} = \{(\check{k}, p) : k < \omega,\ p \in A_k\}, \qquad A_k \subseteq D_{\mathbb{H}} \text{ an antichain.}$$

These names form a fixed set. Names for sets of reals are represented by families of antichains indexed by those real names; the same construction, applied to graphs, supplies names for choice functions. Thus the presentations together with the names and embeddings still form a set, possibly of size greater than $2^{\aleph_1}$. This set includes the restrictions and products in Lemma 2.12. A name for a quotient iteration defined below a condition can be used there or extended by trivial almost disjoint factors on the complement, with the same countable length.

The allowability hierarchy. We define successive hierarchies of allowable iterations. Each hierarchy begins at a specified initial segment of the final iteration. The ξ-th hierarchy keeps the restrictions on coding imposed so far and the fixed classes from hierarchies $\eta < \xi$. A restriction forbids coding a specified triple (m, x, y). At a later extension by a bounded initial segment, evaluate the earlier definitions via the complete embeddings from the initial segments at which those definitions began. In a product with a further iteration these embeddings are obtained through the factor belonging to the final iteration.

For a generic extension M considered in the current hierarchy, let the current class of base forcings $\mathcal{K}_0^{\xi}(M)$ consist of the quotient iterations over M that obey all the restrictions on coding and belong to every earlier fixed class, evaluated over M. Assume now that we have defined $\mathcal{K}_{\beta}^{\xi}$ in all these extensions, put

$$P_{\beta}^{\xi}(M) = \{b \in (\omega \times \mathbb{R}^2)^M : \forall \mathbb{Q} \in \mathcal{K}_{\beta}^{\xi}(M) \quad M \models \mathbb{Q} \Vdash A(b)\}.$$

The rank of b up to (ξ, β) is the first pair, in lexicographic order, at which b belongs to a persistence set. Use all levels of the earlier hierarchies up to

their fixed levels and the levels at most β of the current hierarchy. Write $\rho_{\leq(\xi,\beta)}(b)$ for this partial rank. It is undefined precisely when b has no rank at any of these levels, including those in the earlier hierarchies.

For $m, x \in M$, let $\Lambda^{\xi}_{\beta}(M, m, x)$ be the set of pairs λ for which some $\mathbb{Q} \in \mathcal{K}^{\xi}_{\beta}(M)$, some $q \in \mathbb{Q}$, and some real name $\dot{y}$ satisfy

$$q \Vdash^{M}_{\mathbb{Q}} \rho_{\leq(\xi,\beta)}(m, x, \dot{y}) = \lambda.$$

When this set is nonempty, put $\lambda = \min \Lambda^{\xi}_{\beta}(M, m, x)$. Call this minimum *stable* if no condition witnessing it forces $\dot{y} \notin \mathbb{R}^{M}$. In this case let B be the nonempty set of $y \in \mathbb{R}^{M}$ for which some such condition forces $\rho_{\leq(\xi,\beta)}(m, x, \check{y}) = \lambda$, and put $j^{\xi}_{\beta}(M, m, x) = J_M(B)$. Thus j^{ξ}_{β} is defined precisely in the stable case.

An iteration over M belongs to $\mathcal{K}^{\xi}_{\beta+1}(M)$ if it belongs to $\mathcal{K}^{\xi}_{\beta}(M)$ and, whenever a stage codes (m, x, y), the extension N by the preceding initial segment satisfies

$$j^{\xi}_{\beta}(N, m, x) \downarrow \implies y \neq j^{\xi}_{\beta}(N, m, x). \tag{3}$$

Take intersections at limit levels. At a limit of hierarchies retain the union of the restrictions on coding and use every earlier fixed class in the definition at level zero. We will always be able to produce non-empty classes which form a fixed point in the following way:

Lemma 2.13. *Let M_0 be the model at the start of hierarchy ξ. The preceding recursion uniformly defines $\mathcal{K}^{\xi}_{\beta}(M)$ in every generic extension $M = M_0[G]$ under consideration. There is an ordinal θ_{ξ}, independent of M, such that*

$$\beta \geq \theta_{\xi} \implies \mathcal{K}^{\xi}_{\beta}(M) = \mathcal{K}^{\xi}_{\theta_{\xi}}(M) \qquad \text{for every such } M.$$

Proof. All hypothetical generic filters below are interpreted by the forcing theorem. Induct on β, simultaneously over the chosen forcing presentations. Let $G \subseteq \mathbb{H}$ be M_0-generic and put $M = M_0[G]$. At level zero, retain the restrictions on coding and the earlier fixed classes, interpreted through the specified embeddings.

Suppose level β is defined, and abbreviate $\rho^{N}_{\leq(\xi,\beta)}$ by ρ^{N}_{β}. For $b \in (\omega \times \mathbb{R}^2)^M$, $m \in \omega$, and $x \in \mathbb{R}^M$, the definitions say

$$\begin{aligned} b \in P^{\xi}_{\beta}(M) &\iff \forall \mathbb{Q}, H \quad M[H] \models A(b), \\ \lambda \in \Lambda^{\xi}_{\beta}(M, m, x) &\iff \exists \mathbb{Q}, H, y \quad \rho^{M[H]}_{\beta}(m, x, y) = \lambda, \end{aligned}$$

where $\mathbb{Q} \in \mathcal{K}^{\xi}_{\beta}(M)$, $H \subseteq \mathbb{Q}$ ranges over hypothetical M-generic filters, and $y \in \mathbb{R}^{M[H]}$. At the least attainable λ, stability means that no witness y in the second clause lies outside M. In the stable case, let $B \subseteq \mathbb{R}^M$ be the nonempty set of witnesses from that clause and put $j^{\xi}_{\beta}(M, m, x) = J_M(B)$.

For a candidate iteration $\mathbb{T}$ over M, let $H \subseteq \mathbb{T}$ be hypothetical M-generic. At a coding stage i, write $M_i = M[H_i]$ for the preceding intermediate extension and (m_i, x_i, y_i) for the triple coded there. Then $\mathbb{T} \in \mathcal{K}^{\xi}_{\beta+1}(M)$ iff $\mathbb{T} \in \mathcal{K}^{\xi}_{\beta}(M)$ and, for every such H and every coding stage i,

$$j^{\xi}_{\beta}(M_i, m_i, x_i)\downarrow \quad\Longrightarrow\quad y_i \neq j^{\xi}_{\beta}(M_i, m_i, x_i).$$

At a limit δ, $\mathcal{K}^{\xi}_{\delta}(M) = \bigcap_{\beta<\delta} \mathcal{K}^{\xi}_{\beta}(M)$. The induction hypothesis and the forcing theorem express every clause uniformly by conditions and names in the preceding model.

When restricting or factoring the forcing, use the usual translation of names for all parameters. At a limit of hierarchies, intersect the earlier fixed classes.

Finally, in M_0 fix the set X_ξ of triples $(\mathbb{H}, \dot{\mathbb{T}}, p)$, where $\mathbb{H}, \dot{\mathbb{T}}$ are among the chosen presentations and $p \in \mathbb{H}$, and put

$$D_\beta = \{(\mathbb{H}, \dot{\mathbb{T}}, p) \in X_\xi : p \Vdash_{\mathbb{H}} \dot{\mathbb{T}} \in \mathcal{K}^{\xi}_{\beta}\}.$$

Here the forcing statement uses the interpretation just defined. For $\beta \leq \gamma$, $D_\gamma \subseteq D_\beta$. The least stages at which members of D_0 disappear form a set; choose θ_ξ above all these stages. Then $D_\beta = D_{\theta_\xi}$ for every $\beta \geq \theta_\xi$. For every M_0-generic $G \subseteq \mathbb{H}$, the truth lemma gives

$$\dot{\mathbb{T}}^G \in \mathcal{K}^{\xi}_{\beta}(M_0[G]) \quad\Longleftrightarrow\quad \exists p \in G\ (\mathbb{H}, \dot{\mathbb{T}}, p) \in D_\beta.$$

Since the chosen names cover all relevant tails, the same θ_ξ works in every such generic extension. Put $K_\xi(M) = \mathcal{K}^{\xi}_{\theta_\xi}(M)$. □

Lemma 2.14. *If $\mathbb{Q}, \mathbb{R} \in \mathcal{K}^{\xi}_{\alpha}(M)$, then*

$$(\mathbb{Q} \times \mathbb{R})/G_{\mathbb{Q}} \in \mathcal{K}^{\xi}_{\alpha}(M[G_{\mathbb{Q}}]), \qquad (\mathbb{Q} \times \mathbb{R})/G_{\mathbb{R}} \in \mathcal{K}^{\xi}_{\alpha}(M[G_{\mathbb{R}}]).$$

For $N = M[G_{\mathbb{R}}]$, write ρ^M, ρ^N for the partial ranks up to (ξ, α). Then

$$\rho^N \restriction (\omega \times \mathbb{R}^2)^M = \rho^M, \qquad \Lambda^{\xi}_{\alpha}(N, m, x) = \Lambda^{\xi}_{\alpha}(M, m, x) \quad (m, x \in M).$$

If the common minimum exists, it is stable over M iff it is stable over N. In the stable case the sets of minimizers agree and $j^{\xi}_{\alpha}(N, m, x) = j^{\xi}_{\alpha}(M, m, x)$.

Proof. The proof proceeds by induction lexicographically on (ξ, α), proving the product assertions first. Evaluating the defining requirements in the intermediate generic extensions, as in Lemma 2.13, gives closure under

$$\mathbb{T} \mapsto \mathbb{T} \restriction t, \qquad \mathbb{T} \mapsto \mathbb{T}_i, \qquad \mathbb{T} \mapsto \mathbb{T}/G_{\mathbb{T}_i}.$$

A name given below d is lifted using trivial almost disjoint factors on $\neg d$. Evaluating each requirement immediately before its coding stage also gives

$$\mathbb{T} \in \mathcal{K}^{\xi}_{\alpha}(S), \qquad \mathbb{T} \Vdash \dot{\mathbb{U}} \in \mathcal{K}^{\xi}_{\alpha}(S[G_{\mathbb{T}}]) \\ \Longrightarrow \quad \mathbb{T} * \dot{\mathbb{U}} \in \mathcal{K}^{\xi}_{\alpha}(S).$$

In $\mathbb{Q} \times \mathbb{R}$, each coding stage codes the same real as in its factor, so the coding restrictions hold. Membership in earlier fixed classes follows from the induction on ξ, through the specified embeddings. For $\beta < \alpha$, let S be the extension preceding a stage of one factor and $S' = S[G_{\mathbb{U}}]$ the extension obtained by adding the other factor. By induction,

$$\mathbb{U} \in \mathcal{K}^{\xi}_{\beta}(S), \qquad j^{\xi}_{\beta}(S', m, x) = j^{\xi}_{\beta}(S, m, x),$$

where equality includes definedness. Thus (3) transfers from S to S', proving the product assertions at α.

Fix $N = M[G_{\mathbb{R}}]$ and $b \in (\omega \times \mathbb{R}^2)^M$. We have

$$b \notin P^{\xi}_{\alpha}(N) \quad \Longleftrightarrow \quad \exists \mathbb{T} \in \mathcal{K}^{\xi}_{\alpha}(N)\ \exists t \in \mathbb{T}\ t \Vdash^{N}_{\mathbb{T}} \neg A(b).$$

Given such $\mathbb{T}, t$, choose their names and decide the length below $r \in G_{\mathbb{R}}$. Lift the iteration as above. Then

$$\mathbb{R} * \dot{\mathbb{T}}^{\uparrow} \in \mathcal{K}^{\xi}_{\alpha}(M), \qquad (r, \dot{t}) \Vdash \neg A(b),$$

so $b \notin P^{\xi}_{\alpha}(M)$. Conversely, if $\mathbb{Q} \in \mathcal{K}^{\xi}_{\alpha}(M)$ and $q \Vdash^{M}_{\mathbb{Q}} \neg A(b)$, the witness persists in $\mathbb{Q} \times \mathbb{R}$. Hence, identifying q with its quotient condition,

$$q \Vdash^{N}_{(\mathbb{Q}\times\mathbb{R})/G_{\mathbb{R}}} \neg A(b).$$

Consequently

$$P^{\xi}_{\alpha}(N) \cap M = P^{\xi}_{\alpha}(M).$$

Applying this at the lower levels and earlier hierarchies yields $\rho^N \restriction (\omega \times \mathbb{R}^2)^M = \rho^M$.

Composition $\mathbb{R} * \dot{\mathbb{T}}^{\uparrow}$ and the product $\mathbb{T} \times \mathbb{R}$ give, respectively, the two inclusions in

$$\Lambda^{\xi}_{\alpha}(N, m, x) = \Lambda^{\xi}_{\alpha}(M, m, x).$$

For the product direction, the quotient over $M[G_{\mathbb{T}}]$ belongs to $\mathcal{K}^{\xi}_{\alpha}(M[G_{\mathbb{T}}])$, so the witnessing rank is preserved. For new witnesses, the intersection lemma for mutually generic extensions gives

$$M[G_{\mathbb{T}}] \cap M[G_{\mathbb{R}}] = M, \\ y \in M[G_{\mathbb{T}}] \setminus M \quad \Longrightarrow \quad y \notin N.$$

Conversely, $y \notin N \Rightarrow y \notin M$ in a composed extension. Thus the common minimum is stable over M iff it is stable over N.

In the stable case, put $\lambda = \min \Lambda^{\xi}_{\alpha}(M, m, x)$ and let B_M, B_N be the minimizer sets. Composition gives $B_N \subseteq B_M$, and the product gives $B_M \subseteq B_N$. Rank preservation and the trivial forcing give

$$B_S = \{y \in \mathbb{R}^S : \rho^S(m, x, y) = \lambda\} \qquad (S = M, N).$$

For $B = B_M = B_N$, coherence therefore yields

$$j^{\xi}_{\alpha}(N, m, x) = J_N(B) = J_M(B) = j^{\xi}_{\alpha}(M, m, x).$$

□

In particular, whenever (3) from an earlier level or hierarchy forbids coding a triple over M, the minimum remains stable and the same triple is forbidden by the corresponding requirement at the current fixed level. Indeed, a minimizer already in M witnesses λ using the trivial forcing in the smaller class. Any smaller rank attainable in the smaller class would already have been attained at the earlier level, since newly assigned ranks are later in the lexicographic order. Any new real of rank λ obtainable in the smaller class would also have been obtainable in the earlier class. Thus the set of minimizers and its choice agree. If the current minimum can be attained by a new real, no earlier requirement of this kind is defined at (m, x) over M.

Lemma 2.15. *Let M be an extension by a bounded initial segment of the final iteration during hierarchy η, and let $\mathbb{R} \in K_\eta(M)$ be a further bounded segment of the final iteration. Suppose*

$$(m, x, y) \text{ forbidden during } \mathbb{R} \quad \Longrightarrow \quad y \notin M.$$

Let ξ be the current hierarchy in $M[G_{\mathbb{R}}]$. For every $\mathbb{Q} \in K_\eta(M)$, the quotient $(\mathbb{Q} \times \mathbb{R})/G_{\mathbb{R}}$ belongs to all the intervening fixed classes and to $K_\xi(M[G_{\mathbb{R}}])$. For each initial segment $\mathbb{Q}_i$, put

$$S = M[G_{\mathbb{Q}_i}], \qquad N = M[G_{\mathbb{Q}_i}, G_{\mathbb{R}}].$$

Writing ρ^S, ρ^N for the partial ranks up to (η, θ_η) and (ξ, θ_ξ), respectively, we have

$$\rho^N \restriction (\omega \times \mathbb{R}^2)^S = \rho^S.$$

For $(m, x) \in S$, if the minimum over S in $K_\eta(S)$ exists, then

$$\min \Lambda^{\xi}_{\theta_\xi}(N, m, x) = \min \Lambda^{\eta}_{\theta_\eta}(S, m, x).$$

The two minima have the same stability. In the stable case the sets of minimizers and the choices agree.

Proof. Fix the product presentation and induct on the later hierarchies and their levels, simultaneously for all $\mathbb{Q} \in K_\eta(M)$ and all $\mathbb{Q}_i$. In each bounded extension containing the start of an earlier hierarchy, interpret its definitions using the generic filters induced through the final-iteration factor. Retain this interpretation after a new hierarchy has started. Lemma 2.9(3) retains the embeddings under $\mathbb{Q}_i \mapsto \mathbb{Q}_i * \dot{\mathbb{T}}$, and Lemma 2.12 extends the choice functions coherently.

At level zero, the earlier fixed classes follow from Lemma 2.14 and induction on the hierarchies. A newly forbidden (m, x, y) cannot be coded by $\mathbb{Q}$: at its preceding stage $\mathbb{Q}_i$ we would have

$$y \in M[G_{\mathbb{Q}_i}] \cap M[G_{\mathbb{R}}] = M, \qquad y \notin M.$$

At a limit of hierarchies intersect the earlier fixed classes.

Suppose the product assertion holds at level β of hierarchy ξ. Put

$$S = M[G_{\mathbb{Q}_i}], \qquad N = M[G_{\mathbb{Q}_i}, G_{\mathbb{R}}], \qquad \rho_\beta^N = \rho^N_{\leq(\xi,\beta)}.$$

For $b \in (\omega \times \mathbb{R}^2)^S$,

$$\rho^S(b) \uparrow \implies \exists \mathbb{T} \in K_\eta(S)\ \exists t \in \mathbb{T}\ t \Vdash^S_{\mathbb{T}} \neg A(b).$$

Replace $\mathbb{T}$ by $\mathbb{T} \restriction t$, choose its name and length below a condition in $G_{\mathbb{Q}_i}$, and lift it using trivial almost disjoint factors on the complement. Then

$$\mathbb{Q}_i * \dot{\mathbb{T}}^\uparrow \in K_\eta(M).$$

Extend the specified presentation of $\mathbb{Q}_i \times \mathbb{R}$ to $(\mathbb{Q}_i * \dot{\mathbb{T}}^\uparrow) \times \mathbb{R}$. By induction its appended quotient

$$\mathbb{U} = \big((\mathbb{Q}_i * \dot{\mathbb{T}}^\uparrow) \times \mathbb{R}\big)/(G_{\mathbb{Q}_i} \times G_{\mathbb{R}})$$

satisfies

$$\mathbb{U} \in \mathcal{K}^\xi_\beta(N), \qquad \mathbb{U} \Vdash_N \neg A(b), \qquad b \notin P^\xi_\beta(N).$$

Here the witness to $\neg A(b)$ persists in the product. Previously assigned ranks agree by Lemma 2.14, evaluating the earlier definitions in this same product. Together with the induction,

$$\rho_\beta^N \restriction (\omega \times \mathbb{R}^2)^S = \rho^S.$$

Suppose $\lambda = \min \Lambda^\eta_{\theta_\eta}(S, m, x)$ exists. Let $\mathbb{T} \in K_\eta(S)$ witness λ, restricted below a witnessing condition. Appending $\mathbb{T}$ as above gives

$$\lambda \in \Lambda^\xi_\beta(N, m, x).$$

In the product over S, the intersection lemma gives

$$S[G_{\mathbb{T}}] \cap N = S, \qquad y \in S[G_{\mathbb{T}}] \setminus S \implies y \notin N.$$

Thus a new witness remains new. Conversely, put $\mathbb{V} = (\mathbb{Q}_i \times \mathbb{R})/G_{\mathbb{Q}_i}$. By Lemma 2.14, $\mathbb{V} \in K_\eta(S)$. For a witnessing $\mathbb{U} \in \mathcal{K}^\xi_\beta(N)$, choose and lift its name as above. Then

$$\mathbb{V} * \dot{\mathbb{U}}^\uparrow \in K_\eta(S).$$

Since every rank $\mu < \lambda$ is already a rank up to (η, θ_η), this composition gives

$$\mu < \lambda, \quad \mu \in \Lambda^\xi_\beta(N, m, x) \quad \Longrightarrow \quad \mu \in \Lambda^\eta_{\theta_\eta}(S, m, x),$$

contrary to minimality. Hence

$$\min \Lambda^\xi_\beta(N, m, x) = \lambda.$$

If the old minimum is stable, composition excludes new minimizers over S; the old minimizers retain rank λ. For the corresponding sets B_S, B_N, therefore,

$$B_N \subseteq B_S \subseteq B_N, \qquad j^\xi_\beta(N, m, x) = J_N(B_N) = J_S(B_S) = j^\eta_{\theta_\eta}(S, m, x).$$

Together with the preservation of new witnesses, this proves equality of stability.

At a stage of $\mathbb{Q}$ coding (m, x, y), if the old minimum exists, its stability and choice transfer (3) from S to N. Otherwise, the trivial forcing and rank preservation give

$$\Lambda^\eta_{\theta_\eta}(S, m, x) = \varnothing \implies \rho^S(m, x, y) \uparrow \implies \rho^N_\beta(m, x, y) \uparrow .$$

Thus $j^\xi_\beta(N, m, x) \downarrow$ implies $y \neq j^\xi_\beta(N, m, x)$ in this case as well. This proves the successor step. At limit levels take all earlier requirements, yielding the assertions at the fixed level. □

Lemma 2.16. *Suppose the current fixed class allows its minimum rank λ at (m, x) to be attained by a real outside M. There is a restriction of an allowable iteration that forces a real y of rank λ at (m, x) to appear in its final extension and in no extension by a proper initial segment.*

Proof. Choose $\mathbb{Q} \in K_\xi(M)$, $q \in \mathbb{Q}$, and $\dot{y}$ with

$$q \Vdash_{\mathbb{Q}} \dot{y} \notin \mathbb{R}^M \ \wedge \ \rho_{\leq(\xi,\theta_\xi)}(m, x, \dot{y}) = \lambda.$$

Strengthen q to decide $\delta > 0$ and a $\mathbb{Q}_\delta$-name $\dot{z}$ such that

$$q \Vdash_{\mathbb{Q}} \delta = \min\{\gamma \leq \mathrm{lh}(\mathbb{Q}) : \dot{y} \in \dot{\mathbb{R}}^{\mathbb{Q}_\gamma}\},$$
$$q \Vdash_{\mathbb{Q}} \dot{y} = \dot{z},$$

identifying names with their lifts. Since $\mathbb{Q}/G_{\mathbb{Q}_\delta}$ is allowable, Lemma 2.14 implies that q forces the $\mathbb{Q}_\delta$-extension to satisfy

$$\dot{z} \notin \bigcup_{\gamma<\delta} \dot{\mathbb{R}}^{\mathbb{Q}_\gamma}, \qquad \rho_{\leq(\xi,\theta_\xi)}(m, x, \dot{z}) = \lambda.$$

Take $p \le q$ in the dense suborder and put $p_\delta = \pi_\delta(p)$. Since

$$s \le p_\delta \quad \Longrightarrow \quad s \not\perp_{\mathbb{Q}} p,$$

p_δ forces both assertions over $\mathbb{Q}_\delta$. Thus $\mathbb{Q}_\delta \restriction p_\delta$, with initial segments $\mathbb{Q}_\gamma \restriction \pi_\gamma(p)$, $\gamma \le \delta$, is as required, also when δ is a limit. □

3 The uniformizing extension

Theorem 3.1. *There is a coding iteration $\mathbb{P}$ of length ω_1 over W such that, for every W-generic $G \subseteq \mathbb{P}$, each A_m is uniformized in $W[G]$ by*

$$U_m(x,y) \quad \Longleftrightarrow \quad (x,y) \in A_m \ \wedge \ \neg\Phi(\langle x,y,m\rangle).$$

Proof. We use Lemmas 2.9–2.11 and the absoluteness facts of Subsection 2.6. Let $\mathbb{P}_i$ be the initial segment of length i, and put $G_i = G \cap \mathbb{P}_i$. By Lemma 2.10,

$$\omega_1^{W[G_i]} = \omega_1, \qquad \mathbb{R}^{W[G_i]} = \{x_{i,j} : j < \omega_1\}, \qquad \mathbb{R}^{W[G]} = \bigcup_{i<\omega_1} \mathbb{R}^{W[G_i]},$$

where each enumeration is specified by a $\mathbb{P}_i$-name. If ξ_i is the hierarchy in use at stage i, abbreviate

$$\mathcal{K}_i = K_{\xi_i}(W[G_i]), \qquad \rho_i = \rho^{W[G_i]}_{\le(\xi_i,\theta_{\xi_i})},$$
$$\Lambda_i(m,x) = \Lambda^{\xi_i}_{\theta_{\xi_i}}(W[G_i],m,x), \quad j_i(m,x) = j^{\xi_i}_{\theta_{\xi_i}}(W[G_i],m,x).$$

Let F_ξ be the triples explicitly forbidden in hierarchy ξ. List the following instructions cofinally often, using (i,j) for $x_{i,j}$; skip references to initial segments not yet constructed. Define $y_*(m,x)$ at most once.

1. For $b = (m,x,y) \in W[G_i]$,

   $$\rho_i(b)\!\uparrow \quad \Longrightarrow \quad \exists \mathbb{Q} \in \mathcal{K}_i \ \exists q \in \mathbb{Q} \ q \Vdash^{W[G_i]}_{\mathbb{Q}} \neg A(b).$$

   Append $\mathbb{Q} \restriction q$.

2. If $\mathcal{K}_i$ permits coding (m,x,y), append a coding stage for $\langle x,y,m\rangle$.

3. Suppose $y_*(m,x)\!\uparrow$ and $\Lambda_i(m,x) \neq \varnothing$. Put $\lambda = \min \Lambda_i(m,x)$. If λ is stable, set

   $$y_*(m,x) = j_i(m,x) = J_{W[G_i]}(B),$$

   where B is the set of minimizers. Otherwise, Lemma 2.16 supplies $\mathbb{Q} \in \mathcal{K}_i$, $\delta = \mathrm{lh}(\mathbb{Q}) > 0$, and $\dot y$ such that

   $$\mathbb{Q} \Vdash \dot y \notin \bigcup_{\alpha<\delta} \dot{\mathbb{R}}^{\mathbb{Q}_\alpha} \quad \wedge \quad \rho_{\le(\xi_i,\theta_{\xi_i})}(m,x,\dot y) = \lambda.$$

Append $\mathbb{Q}$, put $y_*(m,x) = \dot{y}^H$ for its generic H, and start a new hierarchy with

$$F_{\xi_i+1} = F_{\xi_i} \cup \{(m, x, y_*(m,x))\}.$$

The restriction applies before the next factor.

The instructions are carried out using the uniformly definable predicates of Lemma 2.13 and the forcing theorem. On a maximal antichain $\mathcal{A} \subseteq \mathbb{P}_i$, decide which instruction applies and choose its witnesses, including $\dot{\mathbb{Q}}_a$ and $\delta_a < \omega_1$ with $a \Vdash \mathrm{lh}(\dot{\mathbb{Q}}_a) = \delta_a$. At the k-th stage within a tail, mix its k-th factor names; at limits take the prescribed limits of initial segments. Thus the length of an instruction may depend on the generic. Between instructions extend the earlier choice functions coherently to all new nonempty sets of reals. At a limit ζ of hierarchies, use

$$F_\zeta = \bigcup_{\xi<\zeta} F_\xi, \qquad \mathcal{K}_0^\zeta(M) = \bigcap_{\xi<\zeta} K_\xi(M).$$

Each new hierarchy starts before the next factor; earlier hierarchies continue to determine previously assigned ranks. If no tail is appended, insert a stage with trivial almost disjoint factor after the choice or the start of a new hierarchy. For the starting stages of the instructions, therefore,

$$\gamma_0 = 0, \qquad \gamma_{\nu+1} = \gamma_\nu + \ell_\nu \quad (0 < \ell_\nu < \omega_1), \qquad \gamma_\delta = \sup_{\nu<\delta} \gamma_\nu \quad (\delta \text{ limit}),$$
$$\nu \le \gamma_\nu < \omega_1, \qquad \sup_{\nu<\omega_1} \gamma_\nu = \omega_1.$$

The last line follows from regularity of ω_1. Together with Lemma 2.10, it ensures that each instruction is considered after its parameters have appeared.

Verification. For $i < j < \omega_1$, the construction gives

$$\mathbb{P}_j/G_i \in \mathcal{K}_i,$$
$$(m,x,y) \in F_{\xi_j} \setminus F_{\xi_i} \quad \Longrightarrow \quad y \notin W[G_i].$$

Thus Lemmas 2.14 and 2.15 yield

$$\rho_j \restriction (\omega \times \mathbb{R}^2)^{W[G_i]} = \rho_i.$$

For $b \in (\omega \times \mathbb{R}^2)^{W[G_i]}$, persistence, the two absoluteness facts and instruction (1) give

$$\begin{aligned} \rho_i(b)\downarrow &\Longrightarrow \forall j \in [i,\omega_1) \quad W[G_j] \models A(b) \\ &\Longrightarrow W[G] \models A(b), \end{aligned}$$

$$\begin{aligned} \rho_i(b)\uparrow &\overset{(1)}{\Longrightarrow} \exists j \in [i,\omega_1) \quad W[G_j] \models \neg A(b) \\ &\Longrightarrow W[G] \models \neg A(b). \end{aligned}$$

Hence

$$W[G] \models A(b) \quad \iff \quad \rho_i(b)\downarrow .$$

Using the trivial forcing and preservation of the least attainable rank,

$$\begin{aligned} A^{W[G]}_{m,x} \neq \varnothing &\implies \exists i, y \quad x, y \in W[G_i] \ \wedge \ \rho_i(m,x,y)\downarrow \\ &\implies \exists i \quad \Lambda_i(m,x) \neq \varnothing \\ &\overset{(3)}{\implies} y_*(m,x)\downarrow \ \wedge W[G] \models A(m,x,y_*(m,x)). \end{aligned}$$

Fix $A^{W[G]}_{m,x} \neq \varnothing$, and let $y_* = y_*(m,x)$ be chosen at γ. Put

$$\lambda = \rho_\gamma(m,x,y_*), \qquad \mathcal{C} = \{u_i : i < \omega_1, \ i \text{ is a coding stage}\}.$$

If y_* was chosen at a stable minimum, preservation gives, for every $i < \omega_1$,

$$\begin{aligned} x, y_* \in W[G_i] &\implies \rho_i(m,x,y_*)\downarrow \\ &\implies \Lambda_i(m,x) \neq \varnothing \\ &\implies j_i(m,x) = j_\gamma(m,x) = y_*. \end{aligned}$$

Therefore (3) gives $\langle x, y_*, m\rangle \notin \mathcal{C}$. Otherwise, instruction (3) gives, also for limit γ,

$$\begin{aligned} i < \gamma \implies y_* \notin W[G_i], \qquad i \geq \gamma &\implies (m,x,y_*) \in F_{\xi_i}, \\ &\implies \langle x, y_*, m\rangle \notin \mathcal{C}. \end{aligned}$$

Let instruction (2) act on (m,x,y) at stage $i \geq \gamma$, where $y \neq y_*$. In the stable case, the observation after Lemma 2.14 gives, for every level (η,β) whose requirement (3) applies at stage i,

$$j^\eta_\beta(W[G_i], m, x)\downarrow \quad \implies \quad j^\eta_\beta(W[G_i], m, x) = y_*.$$

Otherwise, if γ_ν starts the instruction choosing y_*, then $y_* \notin W[G_{\gamma_\nu}]$. By Lemma 2.15 over $W[G_{\gamma_\nu}]$ and the same observation,

$$\begin{aligned} &\exists \mathbb{Q} \in \mathcal{K}_i \ \exists q \in \mathbb{Q} \ \exists \dot{z} \\ &q \Vdash^{W[G_i]}_{\mathbb{Q}} \left(\dot{z} \notin \mathbb{R}^{W[G_i]} \ \wedge \ \rho_{\leq(\xi_i,\theta_{\xi_i})}(m,x,\dot{z}) = \lambda\right), \\ \implies \quad & j^\eta_\beta(W[G_i], m, x)\uparrow \quad \text{for every such level } (\eta,\beta). \end{aligned}$$

Since

$$\{z : (m,x,z) \in F_{\xi_i}\} \subseteq \{y_*\},$$

instruction (2) yields

$$y \neq y_* \quad \implies \quad u_i = \langle x, y, m\rangle \in \mathcal{C}.$$

By Lemma 2.11, in $W[G]$,

$$\Phi(\langle x, y, m\rangle) \iff \langle x, y, m\rangle \in \mathcal{C} \iff y \neq y_*.$$

Together with $U_m \subseteq A_m$, this gives

$$\{y : U_m(x, y)\} = \begin{cases} \{y_*(m, x)\}, & A_{m,x} \neq \varnothing, \\ \varnothing, & A_{m,x} = \varnothing. \end{cases}$$

Finally,

$$A_m \in \Pi^1_k, \quad \Phi \in \Sigma^1_k \implies U_m \in \Pi^1_k.$$

Include real parameters in the first coordinate of a universal relation. □

Remark. The arguments of Lemmas 2.12–2.16 and the construction above use only the iteration and coding properties in Lemmas 2.9–2.11 and the two absoluteness facts of Subsection 2.6. Thus the same argument applies over another ground model once the corresponding properties are verified. We use this in Subsection 4.5.

Corollary 3.2. *There is a cardinal-preserving set-generic extension of L in which* CH *and the Π^1_3-uniformization property hold.*

Proof. First force with the branch product producing W, then with the iteration of Theorem 3.1. Cardinal preservation and CH follow from Theorem 2.7 and Lemma 2.10. Since Φ is Σ^1_3, the uniformizing graphs are Π^1_3. □

4 Forcing over the canonical inner models M_n

Fix for the remainder of this section a natural number n with $1 \leq n < \omega$, and assume that $M_n^{\#}$ exists. As always we let M_n be the resulting inner model one obtains when iterating the sharp out of the universe. Throughout this section, ω_1 denotes $\omega_1^{M_n}$. Our goal is to prove the following theorem.

Theorem. *There is a set-generic extension of M_n in which the Π^1_{n+3}-uniformization property holds.*

The proof follows the construction over L from the preceding sections, but three ingredients require M_n-analogues: the recovery of the canonical lower part $M_n|\omega_1$, the definition of the coding predicate, and the absoluteness argument for witnesses to nonmembership in the sections. We begin with the inner-model-theoretic background needed for these analogues.

4.1 Inner-model-theoretic background and generic absoluteness

We use the notation and comparison conventions of Steel's analysis of mice with finitely many Woodin cardinals; see [19, 20]. An *ordinary premouse* means a premouse in the parameter-free premouse language. In particular, no real or other base set is included as a predicate.

Definition 4.1. *Let $\mathcal{M}$ be a premouse and let $\eta < \mathrm{Ord}^{\mathcal{M}}$. We say that $\mathcal{M}$ is* n-small above η *if, whenever E is an extender on the $\mathcal{M}$-sequence and*

$$\eta < \mathrm{crit}(E),$$

the initial segment $\mathcal{J}^{\mathcal{M}}_{\mathrm{crit}(E)}$ does not have n Woodin cardinals above η. We say that $\mathcal{M}$ is n-small *if it is n-small above 0.*

We use Steel's Π_n-iterability predicates [19, Definitions 1.4 and 1.6]; their definition depends on the parity of n. For a premouse $\mathcal{M}$ and a cutpoint η, write $\mathrm{It}_n(\mathcal{M}, \eta)$ for this predicate above η. We will use the cutpoint ω.

Steel's analysis gives the following complexity statement.

Fact 4.2. *For each fixed $1 \leq n < \omega$, the relation*

"*x codes a countable premouse which is Π_n-iterable above ω.*"

is Π^1_{n+1} in the real code x. Moreover, every countable realizable n-small premouse is Π_n-iterable at the indicated cutpoint.

We next specify the generic extensions in which the comparison lemma will be used. All cardinalities in the following definition are computed in M_n.

Definition 4.3. *A model N is a* small M_n-generic extension *if*

$$N = M_n[G]$$

for some cardinal-preserving forcing $\mathbb{P} \in M_n$ such that $M_n \models |\mathbb{P}| \leq \aleph_1$ and some M_n-generic filter $G \subseteq \mathbb{P}$. A small two-step extension *is an extension*

$$M_n[G] \subseteq M_n[G][H]$$

obtained from a two-step forcing $\mathbb{P} * \dot{\mathbb{Q}} \in M_n$ which is forced to be cardinal preserving and of size at most $\aleph_1$.

The preparatory branch forcing and every forcing iteration used later will be shown to have size $\aleph_1$ and to preserve cardinals. Hence all models which occur in the proof are small M_n-generic extensions in the sense of Definition 4.3.

The second background ingredient is the mouse-operator closure which yields projective generic absoluteness. Recall that a set is *self-wellordered*, abbreviated *swo*, if it codes a wellorder of itself. For a self-wellordered set X, let $M_k^{\#}(X)$ denote the canonical active X-mouse with k Woodin cardinals. Thus $M_0^{\#}(X)$ is the usual sharp over X.

Fact 4.4. *The model M_n is closed under the operator*

$$X \longmapsto M_{n-1}^{\#}(X)$$

below its least Woodin cardinal. In particular, for every self-wellordered $X \in H(\omega_2)^{M_n}$, the mouse $M_{n-1}^{\#}(X)$ exists and is normally ω_2-iterable in M_n.

This closure is preserved by the small forcing notions which occur in the construction.

Lemma 4.5. *Let $N = M_n[G]$ be a small M_n-generic extension. Then*

$$N \models \text{“}M_{n-1}^{\#}(X) \text{ exists and is normally } \omega_2\text{-iterable for every swo } X \in H(\omega_2)\text{.”}$$

Moreover, whenever X belongs to both M_n and N, the two models compute the same mouse $M_{n-1}^{\#}(X)$.

Proof. Put $\kappa = \omega_1^{M_n}$ and $F(X) = M_{n-1}^{\#}(X)$. For every swo $X \in H(\kappa^+)^{M_n}$, Fact 4.4 gives $F(X) \downarrow$ and normal κ^+-iterability. Since $|\mathbb{P}|^{M_n} \leq \kappa$ and $\mathbb{P}$ preserves cardinals, [18, Lemma 3.4], with index $n-1$, gives the same assertion for swo $X \in H(\kappa^+)^N$, as well as $X \in H(\kappa^+)^{M_n}$, X swo $\implies$ $F^N(X) = F^{M_n}(X)$. □

We can now state the precise generic absoluteness principle used later. It is the two-step form which permits real parameters introduced by the first forcing.

Theorem 4.6. *Let*

$$M_n[G] \subseteq M_n[G][H]$$

be a small two-step extension. Then

$$M_n[G] \prec_{\Sigma^1_{n+2}} M_n[G][H].$$

Equivalently, for every real parameter $a \in M_n[G]$ and every Σ^1_{n+2} formula $\vartheta(v)$,

$$M_n[G] \models \vartheta(a) \quad \iff \quad M_n[G][H] \models \vartheta(a).$$

The same equivalence holds for Π^1_{n+2} formulas.

Proof. Put $N_0 = M_n[G]$ and $N_1 = N_0[H]$. By Lemma 4.5, N_0 is closed under $X \mapsto M_{n-1}^{\#}(X)$, with normal ω_2-iterability, for swo $X \in H(\omega_2)^{N_0}$. With $\kappa = \omega_1$ and $|\mathbb{Q}|^{N_0} \leq \kappa$, [18, Lemma 3.7] at index $n-1$ gives $N_0 \prec_{\Sigma^1_{(n-1)+3}} N_1$, $\Sigma^1_{(n-1)+3} = \Sigma^1_{n+2}$. For $\psi \in \Pi^1_{n+2}$, apply this to $\neg\psi \in \Sigma^1_{n+2}$. □

The point at which Theorem 4.6 enters the uniformization proof is recorded in the following consequence.

Corollary 4.7. *Let*

$$M_n[G] \subseteq M_n[G][H]$$

be a small two-step extension, let $a \in M_n[G]$ be a real, and let $\varphi(v)$ be a Σ^1_{n+3} formula. If

$$M_n[G] \models \varphi(a),$$

then

$$M_n[G][H] \models \varphi(a).$$

Consequently, if $A \subseteq (2^\omega)^2$ is Π^1_{n+3} and

$$M_n[G] \models (x,y) \notin A,$$

then $(x,y) \notin A$ remains true in $M_n[G][H]$.

Proof. Put $N_0 = M_n[G]$, $N_1 = M_n[G][H]$, and write $\varphi(a) \equiv \exists z\, \psi(z,a)$, where $\psi \in \Pi^1_{n+2}$. By Theorem 4.6,

$$\begin{aligned} N_0 \models \varphi(a) &\Longrightarrow \exists z \in \mathbb{R}^{N_0}\ (N_0 \models \psi(z,a)) \\ &\Longrightarrow \exists z \in \mathbb{R}^{N_0}\ (N_1 \models \psi(z,a)) \\ &\Longrightarrow N_1 \models \varphi(a). \end{aligned}$$

For the last assertion take $\varphi(x,y) \equiv (x,y) \notin A$. □

4.2 The definable lower part of M_n

Definition 4.8. *Let $\mathcal{I}_n$ be the set of real codes for well-founded ordinary premice P such that*

$$\begin{aligned} P \text{ is } \omega\text{-sound}, &\qquad \rho_\omega(P) = \omega, \\ P \text{ is } n\text{-small}, &\qquad \mathrm{It}_n(P,\omega). \end{aligned}$$

Here ω is a cutpoint, and we use Steel's iterability predicate from [19, Definitions 1.4 and 1.6]. We also write $P \in \mathcal{I}_n$ when a real code for P belongs to this set. For a model N, let $\mathcal{C}_n^N$ be the class of transitive premice which have a code in N satisfying this definition in N.

The definition of $\mathcal{I}_n$ is parameter-free Π^1_{n+1} and is invariant under a change of the enumeration of the coded premouse. Indeed the iterability predicate is Π_n over HC by [19, Lemmas 1.5 and 1.7], giving the bound in real codes in Fact 4.2. The remaining checks use the satisfaction relations of the coded structures and the finite-degree fine-structural definitions. They do not increase the bound, and well-foundedness is Π^1_1.

Lemma 4.9. *Every member of $\mathcal{C}_n^{M_n}$ is an initial segment of M_n of height below ω_1, and their heights are cofinal in ω_1. If N is a small M_n-generic extension, then*

$$\mathcal{C}_n^N = \mathcal{C}_n^{M_n}, \qquad \bigcup \mathcal{C}_n^N = M_n|\omega_1.$$

Proof. Write $M = M_n$. By [19, Theorem 2.3] and [20, Corollaries 3.12–3.13], there is a family $\mathcal{R}$ of countable realizable sound segments $Q \triangleleft M$ such that

$$\rho_\omega(Q) = \omega \quad (Q \in \mathcal{R}), \qquad \mathbb{R}^M = \bigcup_{Q \in \mathcal{R}} (\mathbb{R} \cap Q), \qquad \sup_{Q \in \mathcal{R}} \mathrm{Ord}(Q) = \omega_1.$$

Indeed, a bound $\beta < \omega_1$ on the heights would give $\mathbb{R}^M \subseteq M|\beta$ with $|M|\beta|^M \leq \aleph_0$. By [19, Corollary 4.4], M is Σ_n^{HC}-correct. Hence, for $Q \in \mathcal{R}$, [19, Lemma 2.2(2)] gives

$$V \models \mathrm{It}_n(Q, \omega) \quad \iff \quad M \models \mathrm{It}_n(Q, \omega), \qquad \mathcal{R} \subseteq \mathcal{C}_n^M.$$

For $P \in \mathcal{C}_n^M$, choose $Q \in \mathcal{R}$ with $\mathrm{Ord}(P) < \mathrm{Ord}(Q)$. These are ω-mice agreeing below the cutpoint ω, so [19, Lemma 2.2(3)] yields $P \trianglelefteq Q \ \vee \ Q \trianglelefteq P \quad \Longrightarrow \quad P \trianglelefteq Q \triangleleft M$. Let N be a small extension. For premouse codes p, q with coded structures P_p, P_q, put

$$\mathrm{Comp}(p, q) \iff (\exists Q \trianglelefteq P_q)\ P_p \cong Q \ \vee \ (\exists Q \trianglelefteq P_p)\ P_q \cong Q.$$

An initial-segment cut and an isomorphism have real codes; thus $\mathrm{Comp} \in \Sigma^1_1$. If $q \in M$ codes a member of $\mathcal{C}_n^M$, then

$$\chi(q) \equiv \exists p\, (p \in \mathcal{I}_n \wedge \neg\, \mathrm{Comp}(p, q)) \ \in \Sigma^1_{n+2}.$$

The comparison above gives $M \models \neg\chi(q)$, and Theorem 4.6, with trivial first forcing, gives $N \models \neg\chi(q)$.

For $P \in \mathcal{C}_n^N$, preservation of ω_1 gives $\mathrm{Ord}(P) < \omega_1$. Choose $Q \in \mathcal{R}$ above this height and a code $q \in M$ for Q. Then $N \models \neg\chi(q)$ implies $P \trianglelefteq Q$. Consequently P has a code $p_0 \in M$, and isomorphism invariance followed by absoluteness gives $N \models p_0 \in \mathcal{I}_n \quad \iff \quad M \models p_0 \in \mathcal{I}_n$. Thus $\mathcal{C}_n^N \subseteq \mathcal{C}_n^M$. The reverse inclusion follows by the same absoluteness applied to each ground-model code. Finally,

$$\mathcal{C}_n^N = \mathcal{C}_n^M, \qquad \bigcup \mathcal{C}_n^N = \bigcup \mathcal{R} = M|\omega_1.$$

□

Canonical trees. Write $M = M_n$ and use $<_M$ for its canonical construction order. The order on any initial segment is its own construction order, and every predecessor of an element of that segment belongs to it [19, 20]. All the following choices are made in M.

Lemma 4.10. *There is a canonical $\diamondsuit_{\omega_1}$ sequence $\langle a_\lambda : \lambda < \omega_1\rangle$ over M such that, for every $\lambda < \omega_1$, all initial segments of M containing a computation of $\langle a_\gamma : \gamma < \lambda\rangle$ compute the same sequence.*

Proof. Put $a_0 = a_{\eta+1} = \varnothing$. At a nonzero limit $\lambda < \omega_1$, define

$$(X_\lambda, C_\lambda) = \min_{<_M}\{(X, C) : X \subseteq \lambda,\ C \subseteq \lambda \text{ club}, \\ \forall \gamma \in C\ (a_\gamma \neq X \cap \gamma)\}, \qquad a_\lambda = X_\lambda.$$

The set of candidates is nonempty: choose $\gamma_j \nearrow \lambda$ with $j < \gamma_j$, and take $C = \{\gamma_j : j < \omega\}$, $\quad X = \{j < \omega : j \notin a_{\gamma_j}\}$. Then $j \in X \cap \gamma_j \iff j \notin a_{\gamma_j}$. The condition on (X, C) refers only to the previous values and to λ, X, C. Since $<_M$ restricts to the construction order of each initial segment, any segment containing the preceding computation and its least witness makes the same choice.

For diamond, use Jensen's condensation argument in M [20, Theorems 5.1 and 5.4(1)]. Suppose (X, C) is the $<_M$-least counterexample at ω_1. Choose an ω-sound $Q \triangleleft M$ with

$$X, C, \langle a_\lambda : \lambda < \omega_1\rangle \in Q, \qquad \rho_\omega(Q) = \omega_1^Q = \omega_1.$$

For this projectum calculation, new subsets of ω_1 appear cofinally below $M|\omega_1^+$, supplying sound levels with $\rho_\omega(Q) \leq \omega_1$; see [20, Sections 2 and 5]. If $\rho_\omega(Q) < \omega_1$, its hull together with the standard parameters is countable in M, contrary to $\omega_1 \in Q$.

Take a countable elementary hull $H \prec Q$ containing the displayed data and the standard parameters through a degree witnessing $\rho_\omega(Q) = \omega_1$. For its inverse collapse π,

$$\pi : \bar{Q} \cong H \prec Q, \qquad \delta = H \cap \omega_1 = \mathrm{crit}(\pi) = \rho_\omega(\bar{Q}).$$

The second alternative of the condensation lemma, which involves an extender, does not occur. For the corresponding degree-zero ultrapower j, its critical point $\nu < \delta$ is countable in Q. A surjection $s : \omega \twoheadrightarrow \nu$ in Q would satisfy

$$j(s)(k) = j(s(k)) = s(k) \quad (k < \omega), \qquad j(\nu) = \mathrm{ran}(j(s)) = \mathrm{ran}(s) = \nu,$$

a contradiction. Hence $\bar{Q} \triangleleft M$.

In $\bar{Q}$, the pair $(X \cap \delta, C \cap \delta)$ is least for the recursion below δ. The condition is absolute, and the construction order of $\bar{Q}$ is an initial segment of $<_M$, so this pair is least in M as well. Therefore $a_\delta = X \cap \delta, \qquad \delta \in C, \qquad a_\delta \neq X \cap \delta,$ the required contradiction. □

Use this diamond in the simultaneous construction of Lemma 2.5, with the $<_M$-least choice at every step. It gives independent normal splitting Suslin trees $\vec{S}^n = \langle S^n_\xi : \xi < \omega_1 \rangle$ in M. In particular, the finite product argument there seals each guessed maximal antichain in the appropriate product. At each level use the $<_M$-least enumeration by an initial segment of ω to label its nodes. These labels are used in the coding below.

4.3 The preparatory branch extension W_n

We now pass to the ground model over which the coding iteration will be performed. Let

$$\vec{S}^n = \langle S^n_\xi \mid \xi < \omega_1 \rangle$$

be the independent sequence just constructed. All products in this subsection are computed in M_n.

Definition 4.11. *For $A \subseteq \omega_1$ with $A \in M_n$, put*

$$\mathrm{Br}^n_A := \prod^{\mathrm{fs}}_{\xi \in A} \mathbb{S}^n_\xi.$$

We write

$$\mathrm{Br}_n := \mathrm{Br}^n_{\omega_1} = \prod^{\mathrm{fs}}_{\xi < \omega_1} \mathbb{S}^n_\xi.$$

Let $H \subseteq \mathrm{Br}_n$ be M_n-generic. For $\xi < \omega_1$, define

$$b_\xi := \bigcup \{ p(\xi) : p \in H \text{ and } \xi \in \mathrm{dom}(p) \}.$$

For $A \subseteq \omega_1$ in M_n, let

$$H_A := H \cap \mathrm{Br}^n_A \,.$$

Finally, put

$$W_n := M_n[H].$$

Thus b_ξ is the branch added through S^n_ξ by the ξ-th coordinate. As before, to *destroy* a tree S^n_ξ means to add a cofinal generic branch through it.

Lemma 4.12. *Let $A, B \in M_n$ satisfy*

$$A \subseteq B \subseteq \omega_1.$$

Then the following hold in M_n.

1. *Br^n_A is c.c.c.*
2. *Br^n_A is a complete subforcing of Br^n_B, and*

$$\mathrm{Br}^n_B \cong \mathrm{Br}^n_A \times \mathrm{Br}^n_{B \setminus A} \,.$$

3. *If $\xi \notin A$, then*

$$\mathrm{Br}^n_A \Vdash \text{“}S^n_\xi \text{ is a Suslin tree.”}$$

Proof. Independence gives $\prod_{\xi\in e} S^n_\xi$ Suslin for $\varnothing \neq e \in [\omega_1]^{<\omega}$ in M_n. Apply Lemma 2.2 there for (1). The coordinate map $p \longmapsto (p \restriction A, p \restriction (B \setminus A))$ gives (2), including $\mathrm{Br}^n_A \lessdot \mathrm{Br}^n_B$. For $\xi \notin A$,

$$\mathrm{Br}^n_A \times \mathbb{S}^n_\xi \cong \mathrm{Br}^n_{A\cup\{\xi\}} \text{ is c.c.c.} \quad\Longrightarrow\quad \mathrm{Br}^n_A \Vdash \text{“}S^n_\xi \text{ is Suslin,”}$$

by the product characterization of Suslin-tree preservation. □

Theorem 4.13. *The model $W_n = M_n[H]$ has the following properties.*

1. *The forcing Br_n is c.c.c. and has size $\aleph_1$ in M_n. Consequently, W_n preserves every cardinal and cofinality of M_n, and*

$$W_n \models \mathsf{CH}.$$

2. *Every S^n_ξ is destroyed in W_n, as witnessed by the branch b_ξ.*

3. *If $A \subseteq \omega_1$ belongs to M_n, then H_A is Br^n_A-generic over M_n, and, for every $\xi < \omega_1$,*

$$M_n[H_A] \models \text{“}S^n_\xi \text{ has a cofinal branch”} \quad\Longleftrightarrow\quad \xi \in A.$$

4. *Every real in W_n belongs to $M_n[H_A]$ for some countable set $A \subseteq \omega_1$ with $A \in M_n$.*

5. *More generally, every member of $H(\omega_1)^{W_n}$ belongs to $M_n[H_A]$ for some countable set $A \subseteq \omega_1$ with $A \in M_n$.*

Proof. Apply Theorem 2.7 inside M_n, with

$$(L, \vec{S}, \mathrm{Br}, W) \quad\mapsto\quad (M_n, \vec{S}^{\,n}, \mathrm{Br}_n, W_n).$$

Lemma 4.12 supplies the same product and preservation hypotheses; the argument with nice names from the proof of Theorem 2.7 gives (4) and (5). □

The last two clauses are the branch-coordinate localization needed by the coding argument. We record a form in which a prescribed countable set of coordinates may simultaneously be retained.

Corollary 4.14. *Let $a \in H(\omega_1)^{W_n}$, and let $C \in [\omega_1]^\omega \cap M_n$. Then there is $A \in [\omega_1]^\omega \cap M_n$ such that*

$$C \subseteq A, \qquad a \in M_n[H_A],$$

and

$$M_n[H_A] \models \text{“}S^n_\xi \text{ is a Suslin tree”} \qquad (\xi \in \omega_1 \setminus A).$$

In particular, no cofinal branch through any S^n_ξ with $\xi \notin A$ belongs to $M_n[H_A]$.

Proof. By Theorem 4.13(5), choose $A_0 \in [\omega_1]^\omega \cap M_n$ with $a \in M_n[H_{A_0}]$. Put $A = A_0 \cup C \in [\omega_1]^\omega \cap M_n$. Then $\mathrm{Br}^n_{A_0} \lessdot \mathrm{Br}^n_A$, $\quad a \in M_n[H_{A_0}] \subseteq M_n[H_A]$. For $\xi \notin A$, Lemma 4.12(3) gives $M_n[H_A] \models$ "S^n_ξ is Suslin," hence no cofinal branch through S^n_ξ in this model. □

4.4 The coding over M_n

Write $m^n_\infty = M_n|\omega_1$ and use its fixed ordinal code, with all premouse predicates, in $L[m^n_\infty, r]$. This code is in M_n, and its transitive collapse recovers the lower part.

The almost disjoint family. Fix a recursive injection $\#\colon 2^{<\omega} \to \omega$. In M define, for $\xi < \omega_1$,

$$x_\xi = \min_{<_M}\bigl(2^\omega \setminus \{x_\zeta : \zeta < \xi\}\bigr), \qquad d_\xi = \{\#(x_\xi \restriction k) : k < \omega\}.$$

Here $<_M$ is the canonical construction order on M. Each step is defined because ξ is countable and $M \models \mathsf{CH}$. The set d_ξ is infinite, and if $\zeta < \xi$ and k is the first position at which x_ζ and x_ξ differ, then

$$d_\zeta \cap d_\xi = \{\#(x_\xi \restriction j) : j \le k\}.$$

Thus $D^n = \langle d_\xi : \xi < \omega_1\rangle$ is almost disjoint. Write $D = D^n$ throughout this subsection. Its initial segments are computed in initial segments of M below ω_1, as shown below.

For $Y \subseteq \omega_1$, the conditions of $\mathbb{A}^n(Y)$ are pairs $(s, F) \in [\omega]^{<\omega} \times [Y]^{<\omega}$, ordered by

$$(t, G) \le (s, F) \quad\Longleftrightarrow\quad s \subseteq t, \quad F \subseteq G, \quad (t \setminus s) \cap \bigcup_{\xi \in F} d_\xi = \varnothing.$$

For each stem s, the set $\{(s, F) : F \in [Y]^{<\omega}\}$ is centered, since $(s, \bigcup_{j<\ell} F_j) \le (s, F_j) \quad (j < \ell)$. For a generic filter G let $r = \bigcup\{s : (s, F) \in G\}$. Density gives

$$\xi \in Y \Longrightarrow \exists (s, F) \in G\ (\xi \in F) \Longrightarrow |r \cap d_\xi| < \omega,$$
$$\xi \notin Y \Longrightarrow |d_\xi \setminus \textstyle\bigcup_{\zeta \in F} d_\zeta| = \aleph_0 \Longrightarrow |r \cap d_\xi| = \aleph_0.$$

In the second line, arbitrarily many such points can be added to any stem. Thus $Y = \{\xi < \omega_1 : |r \cap d_\xi| < \omega\}$.

The coding sets. Fix in M a bijection $\varrho_n\colon ({}^{<\omega_1}2)^M \to \omega_1$, using CH in M. Let $\mathbb{C}_{M_n} = ({}^{<\omega_1}2)^M$, ordered by end-extension. This is a dense suborder of $\mathrm{Add}(\omega_1, 1)^M$. For a $\mathbb{C}_{M_n}$-generic g over the current model set

$$h(g) = \{\varrho_n(g \restriction \eta) : \eta < \omega_1\},$$
$$\mathrm{Sel}(h, u) = \{\tau_u(\gamma, k) : \gamma \in h,\ k < \omega\}, \qquad \tau_u(\gamma, k) = \omega\gamma + 2k + u(k).$$

We call $h(g)$ the *coding set* of g. Its cardinality is ω_1. For distinct g and g' the intersection $h(g) \cap h(g')$ is bounded in ω_1, since it consists of the images of their common initial segments.

To code $u \in 2^\omega$, let g be the generic subset of ω_1 added at a new Cohen coordinate and put $h = h(g)$. The forcing at the second step is $\mathbb{A}^n(Y_n(u,g))$, where $Y_n(u,g) \subseteq \omega_1$ is defined below. The set $Y_n(u,g)$ codes u, h, and the branches b_ξ, $\xi \in \mathrm{Sel}(h,u)$.

Codes and restriction to countable heights. Fix a finite fragment of ZFC^- containing the instances of separation, collection and replacement used below, the existence of satisfaction relations for set-sized structures and of transitive collapses of wellorders, and add the axiom that ω_1 exists. A *decoding model* is a countable transitive model of this theory. Let $\langle\cdot,\cdot\rangle_{\mathrm{ord}}$ be Gödel's pairing function on ordinals, which enumerates pairs by increasing maximum and lexicographically for a fixed maximum; for triples iterate it. If N is a decoding model and $\alpha = \omega_1^N$, then pairing maps $\alpha \times \alpha$ onto α. Indeed, the part of the enumeration up to any pair of ordinals below α is countable in N. Conversely the rank of a pair in the enumeration is at least the maximum of its entries. Similarly, $j \cdot \zeta + i < \alpha$ if and only if $\zeta < \alpha$, for $i < j < \omega$. Also $\tau_u(\gamma,k) < \alpha$ for $\gamma < \alpha$ and $k < \omega$, since $\alpha = \omega_1^N$.

Present the nodes of each S^n_ξ by pairs (λ, i), where λ is the level and i is the index in the $<_M$-least bijection from an initial segment of ω onto that level. The bijection is chosen as soon as the level has been constructed. We use these relabelled trees throughout, with the tree orders transported along the bijections. In the codes below a node is identified with the ordinal pairing of λ, i. Thus the restriction of the indexed tree to level $< \alpha$ is obtained by keeping exactly the node indices below α. We regard each b_ξ as the set of indices of the nodes of its downward closure.

Localization. For $\gamma \in h$ and $k < \omega$ use the index $\tau_u(\gamma,k)$ and put

$$B = \{\langle \gamma, k, \zeta\rangle_{\mathrm{ord}} : \gamma \in h,\ k < \omega,\ \zeta \in b_{\tau_u(\gamma,k)}\},$$
$$X = \{3\cdot k : k < \omega,\ u(k) = 1\} \cup \{3\cdot\gamma + 1 : \gamma \in h\} \cup \{3\cdot z + 2 : z \in B\}.$$

The odd part of $Y_n(u,g)$ codes X. For the even part, define a continuous increasing sequence in $\mathrm{acc}(h)$ by

$$c_0 = \min\{\zeta \in \mathrm{acc}(h) : \omega < \zeta\},$$
$$c_{\nu+1} = \min\{\zeta \in \mathrm{acc}(h) : c_\nu + \omega < \zeta\}, \quad c_\lambda = \sup_{\nu<\lambda} c_\nu \quad (\lambda \text{ a nonzero limit}).$$

Fix in M a family $\langle w_\eta : \omega \le \eta < \omega_1\rangle$ with w_η coding a wellorder of ω of type η. Define

$$E = w_{c_0} \cup \{c_\nu + k : \nu < \omega_1,\ k \in w_{c_{\nu+1}}\},$$
$$Y_n(u,g) = \{2\cdot\zeta : \zeta \in E\} \cup \{2\cdot\zeta + 1 : \zeta \in X\}.$$

Here the wellorder codes use the fixed recursive coding of relations on ω. As in the construction over L, the even part localizes the code, and the odd part codes the branch data. The set $Y_n(u, g)$ is defined from g, u, the selected branches and parameters in M.

Suppose N is a decoding model, $\alpha = \omega_1^N$, and $Y_n(u, g) \cap \alpha \in N$. Put $C = \{c_\nu : \nu < \omega_1\}$; continuity and strict increase make C club. If $\alpha \notin C$, the even part yields

case	block of the even part	decoded wellorder
$\alpha < c_0$	$[0, \omega)$	w_{c_0}
$c_\nu < \alpha < c_{\nu+1}$	$[c_\nu, c_\nu + \omega)$	$w_{c_{\nu+1}}$

In the second case, $c_\nu \in N$ is countable there, so $c_\nu + \omega < \alpha$. The decoded relation is well-founded in the ambient model, hence in N. Its collapse in N would give, for $\eta = c_0$ or $c_{\nu+1}$, $f \in N$, $\quad f : (\omega, w_\eta) \cong \beta$, $\quad \beta < \alpha < \eta$. The same f is an isomorphism in the ambient model, so $\beta = \eta$, a contradiction. These cases exhaust $\omega_1 \setminus C$. Thus $\alpha \in C \subseteq \mathrm{acc}(h)$, $\qquad \sup(h \cap \alpha) = \alpha$. Using the ordinal pairing and the node labels, N decodes from the odd part

$$(u, h \cap \alpha, \langle b_{\tau_u(\gamma,k)} \restriction \alpha : \gamma \in h \cap \alpha,\ k < \omega \rangle).$$

Each restricted branch has exactly one node at every level $\lambda < \alpha$.

Correct local sequences. The set $\mathcal{I}_n$ is given by Definition 4.8. By Lemma 4.9, the class $\mathcal{C}_n^N$ is the same for all small extensions N used here. For each $\alpha < \omega_1$ put

$$D^\alpha = D \restriction \alpha, \qquad \vec{S}^{\,n,\alpha} = \langle S_\xi^n \restriction \alpha : \xi < \alpha \rangle,$$

where the latter restriction is by levels. A triple $(m, D', \vec{S}')$ consists of a nonempty well-founded premouse of height $\alpha = \mathrm{Ord}(m)$ which is the union of its proper initial segments, together with sequences of the indicated shape. The triple is *correct* if for every $\tau < \alpha$ there is $\beta \in (\tau, \alpha)$ such that $m|\beta \in \mathcal{I}_n$ and a computation in $m|\beta$ agrees with $D', \vec{S}'$ below τ. This means agreement on the entries of D' of index $< \tau$ and on the trees of index $< \tau$ restricted to levels $< \tau$, including node indices and orders. A *computation* is an element of $m|\beta$ recording an initial part of the constructions of D and $\vec{S}^{\,n}$.

We show that correct triples agree with M and that they exist at club many heights. For every initial segment $M|\beta$,

$$<_{M|\beta} = <_M \restriction (M|\beta)^2, \qquad x \in M|\beta \implies \{y : y <_M x\} \subseteq M|\beta.$$

A computation of D records x_ξ, d_ξ and checks

$$x_\xi \in 2^\omega \setminus \{x_\zeta : \zeta < \xi\}, \qquad d_\xi = \{\#(x_\xi \restriction j) : j < \omega\},$$
$$\forall y \in 2^\omega\, (y <_M x_\xi \Rightarrow \exists \zeta < \xi\ (y = x_\zeta)).$$

The quantifier ranges over the segment containing the computation. All earlier candidates belong to that segment, so induction on ξ gives its agreement with M. For the diamond recursion, record the least-pair witnesses from Lemma 4.10; the conditions on them are absolute once the preceding values are fixed, and the same induction applies.

A computation of the trees records the orders below the current level, the antichains guessed by the diamond sequence, the branches at limit levels, and the level bijections. If a_λ codes a maximal antichain A in the product on a finite set e, restricted below λ, the chosen branches must satisfy

$$\forall f \in {}^e\omega \; \exists t \in A \; \forall \xi \in e \; (t(\xi) \in b_{\xi, f(\xi)}).$$

The branches for each tree must also cover all its preceding nodes. These requirements refer only to the countable structures already constructed, and witnesses exist by Lemma 2.5. Hence a computation and its least witness in $M|\beta$ give the same choice in M: $\min_{<_{M|\beta}}\{b \in M|\beta : T(b)\} = \min_{<_M}\{b \in M : T(b)\}$, where T is the condition on the choice, with the preceding data fixed.

Each computation up to a countable stage is hereditarily countable in M, so [20, Section 2] and Lemma 4.9 give computation up to $\tau \in H^M_{\omega_1} = M|\omega_1 = \bigcup \mathcal{C}_n^{M_n}$. Choose $f \in M$, $f : \omega_1 \to \omega_1$, such that $\tau < f(\tau)$, $\quad M|f(\tau) \in \mathcal{I}_n$, with the computations up to τ belonging to $M|f(\tau)$. Well-foundedness is Π^1_1; the computation checks are arithmetic, and all quantifiers over segments and computations range over countable codes. Thus the predicate $\mathrm{Correct}_n$ expressing correctness is Π^1_{n+1}, and

$$\mathrm{Correct}_n(m, D', \vec{S}') \implies (m, D', \vec{S}') = (M|\alpha, D^\alpha, \vec{S}^{\,n,\alpha}).$$

In the current small extension choose H_Θ sufficiently large and a countable $A \prec H_\Theta$ containing the parameters in use, $m^n_\infty, \omega_1, D, \vec{S}^{\,n}, f$, the map $\lambda \mapsto M|\lambda$, and enumerations in M of each $M|\lambda$, $0 < \lambda < \omega_1$. For the collapse $\pi : A \cong N$ and $\alpha = A \cap \omega_1$, these enumerations give

$$A \cap m^n_\infty = M|\alpha, \qquad \omega_1^N = \alpha, \qquad f\text{“}\alpha \subseteq \alpha,$$
$$\pi(m^n_\infty) = M|\alpha, \qquad \pi(D) = D^\alpha, \qquad \pi(\vec{S}^{\,n}) = \vec{S}^{\,n,\alpha}.$$

For $\xi < \alpha$, the node indices of S^n_ξ in A are exactly those below α; the collapse preserves them and their order. Closure under f supplies the computations required for correctness. A continuous cofinal chain of these submodels gives a club of such heights α.

Lemma 4.15. *There are parameter-free formulas*

$$\Psi_n(r, u) \in \Pi^1_{n+2}, \qquad \Phi_n(u) \iff \exists r \, \Psi_n(r, u), \qquad \Phi_n \in \Sigma^1_{n+3},$$

with the following properties.

1. *If r is the real added at a coding stage for u, then $\Psi_n(r,u)$ holds in the resulting extension and in every further small extension.*

2. *If $\Psi_n(r,u)$ holds in a small extension, then $u \in L[m_\infty^n, r]$, and there is an unbounded $h \subseteq \omega_1$ in $L[m_\infty^n, r]$ such that $L[m_\infty^n, r]$ contains a cofinal branch through S_ξ^n for every $\xi \in \mathrm{Sel}(h,u)$.*

Proof. Let $\mathrm{Dec}_n(c;r,u)$ assert that c codes a decoding model N with $r, u, (m, D', \vec{S}') \in N$, $\mathrm{Ord}(m) = \omega_1^N = \alpha$, and $(m, D', \vec{S}')$ correct in the ambient model. In N, decode (u', h', B') from the odd part of

$$Y_r^N = \{\xi < \alpha : |r \cap d'_\xi| < \omega\}, \qquad B'_{\gamma,k} = \{\zeta : \langle \gamma, k, \zeta\rangle_{\mathrm{ord}} \in B'\}.$$

Write $\mathrm{Br}_\alpha(b,S)$ for the assertion that $b \subseteq S$ is linearly ordered by $<_S$ and $\forall \lambda < \alpha\ (|b \cap S_\lambda| = 1)$. Define

$$\begin{aligned}\mathrm{Test}(c;r,u) \iff\ & u' = u \ \wedge\ \sup h' = \alpha \\ & \wedge\ \forall \gamma \in h'\ \forall k < \omega\ \ \mathrm{Br}_\alpha(B'_{\gamma,k}, S'_{\tau_u(\gamma,k)}).\end{aligned}$$

Include satisfaction relations in c, checked by the arithmetic Tarski clauses. Since $\mathrm{Dec}_n \in \Pi^1_{n+1}$ and Test is arithmetic,

$$\Psi_n(r,u) \equiv \forall c\,(\mathrm{Dec}_n(c;r,u) \Rightarrow \mathrm{Test}(c;r,u)) \ \in \Pi^1_{n+2}.$$

Thus $\Phi_n(u) \equiv \exists r\, \Psi_n(r,u)$ is parameter-free Σ^1_{n+3}.

Suppose r is added by $\mathbb{A}^n(Y_n(u,g))$, possibly restricted below a condition. Put $Y_r = \{\xi < \omega_1 : |r \cap d_\xi| < \omega\}$. For every c satisfying $\mathrm{Dec}_n(c;r,u)$, correctness and localization yield

$$\begin{aligned} Y_r = Y_n(u,g), \qquad & Y_r^N = Y_r \cap \alpha, \qquad \alpha \in \mathrm{acc}(h), \\ (u', h', B') = (u, h \cap \alpha, B \cap \alpha), \qquad & B'_{\gamma,k} = b_{\tau_u(\gamma,k)} \restriction \alpha \quad (\gamma \in h \cap \alpha,\ k < \omega). \end{aligned}$$

Hence $\mathrm{Test}(c;r,u)$ and therefore $\Psi_n(r,u)$. For any further small extension $N_0 \subseteq N_1$, Theorem 4.6 gives $N_0 \models \Psi_n(r,u) \iff N_1 \models \Psi_n(r,u)$.

Conversely, assume $\Psi_n(r,u)$. In $L[m_\infty^n, r]$, the coherent computations in m_∞^n, with bounds cofinal in ω_1, recover $D, \vec{S}^n$. Decode $(\hat{u}, \hat{h}, \hat{B})$ from Y_r. Let $\mathcal{D}_{r,u}$ be the heights ω_1^N of models coded by c with $\mathrm{Dec}_n(c;r,u)$; this set contains a club by the argument above for correct triples. At every $\alpha \in \mathcal{D}_{r,u}$,

$$Y_r^N = Y_r \cap \alpha, \qquad (u', h', B') = (\hat{u}, \hat{h} \cap \alpha, \hat{B} \cap \alpha).$$

The formulas $\mathrm{Test}(c;r,u)$ therefore give $\hat{u} = u$ and $\sup \hat{h} = \omega_1$. Fix $\gamma \in \hat{h}$, $k < \omega$, and $\xi = \tau_u(\gamma, k)$. Then

$$\alpha \in \mathcal{D}_{r,u},\ \alpha > \max\{\gamma, \xi\} \quad \Longrightarrow \quad \mathrm{Br}_\alpha(\hat{B}_{\gamma,k} \cap \alpha, S_\xi^n \restriction \alpha).$$

For any two nodes and any level there is such an α above them, whence $\mathrm{Br}_{\omega_1}(\hat{B}_{\gamma,k}, S_\xi^n)$, with $\hat{B}_{\gamma,k} \in L[m_\infty^n, r]$. $\square$

Coding iterations and exactness

For a set $E \in M_n$ of at most $\aleph_1$ Cohen indices, put

$$\mathbb{C}^n_E = \mathrm{Fn}_{<\omega_1}(E \times \omega_1, 2)^{M_n}.$$

Use the same sets E_i of Cohen indices as in the construction over L, with the reindexing maps now taken in M_n. For $\beta \leq \omega_1$, the forcing through the stages $i < \beta$ is

$$\mathbb{P}^n_\beta = \mathbb{C}^n_{E_{<\beta}} * \dot{\mathbb{R}}^n_\beta, \qquad E_{<\beta} = \bigcup_{i<\beta} E_i, \qquad \mathbb{R}^n_\beta = \operatorname*{fs}_{i<\beta} * \dot{\mathbb{A}}_i.$$

At stage i, names over $\mathbb{P}^n_i$ specify whether to code a real, the real u_i to be coded, the selected coordinate in E_i, and the ground-model isomorphism onto a restriction of $\mathbb{C}^n_{E_i}$. If the stage codes a real, let g_i be the Cohen subset of ω_1 obtained at the selected coordinate after applying this isomorphism. Then $\dot{\mathbb{A}}_i$ is a name for $\mathbb{A}^{\,n}(Y_n(u_i, g_i))$, possibly restricted below a condition named over $\mathbb{P}^n_i * \mathbb{C}^n_{E_i}$. If there is no coding at stage i, its second factor is trivial.

The proof of Lemma 2.9 applies over M_n. For disjoint $E, F \in M_n$,

$$\mathbb{C}^n_{E\cup F} \cong \mathbb{C}^n_E \times \mathbb{C}^n_F, \qquad \pi_{\lambda\beta}(c, s) = \big(c \restriction (E_{<\beta} \times \omega_1), s \restriction \beta\big).$$

Given u_i, g_i, the lower part and selected branches determine $Y_i = Y_n(u_i, g_i)$; the order of the next almost disjoint factor is determined by D^n, Y_i and the condition below which it is restricted. Thus the projections give $\mathbb{P}^n_\beta \lessdot \mathbb{P}^n_\lambda$.

For restrictions, use the ground-model isomorphisms $\psi_c : \mathbb{C}^n_E \cong \mathbb{C}^n_E \restriction c$. They change each coordinate only on a bounded set. Compose them with the stage isomorphisms; the translated data then satisfy $(g'_i, u'_i, Y'_i) = (g_i, u_i, Y_i)$. This yields the presentations below conditions and their commuting initial-segment projections. A tail defined below a Boolean value b is extended by mixing the names of its almost disjoint factors as $\dot{\mathbb{A}}^*_i = \dot{\mathbb{A}}_i$ on b and $\dot{\mathbb{A}}^*_i = \mathbf{1}$ on $\neg b$.

For two tails, reindex their Cohen coordinates disjointly in M_n and translate each factor's names along its initial-segment embeddings. The data g_i, u_i, Y_i agree in either order, giving $\mathbb{R} \times \mathbb{Q} \cong \mathbb{R} * \dot{\mathbb{Q}} \cong \mathbb{Q} * \dot{\mathbb{R}}$. For a prescribed $\mathbb{R} \times \mathbb{Q}_j$ and a continuation $\mathbb{T}$ of $\mathbb{Q}_j$ below b, retain both index maps and extend only the $\mathbb{Q}_j$-map to the new coordinates. Use the image of b under $\mathbb{Q}_j \lessdot \mathbb{R} \times \mathbb{Q}_j$ for the almost disjoint factors of $\mathbb{T}$. Then $\mathbb{R} \times (\mathbb{Q}_j * \mathbb{T}) \cong (\mathbb{R} \times \mathbb{Q}_j) * \dot{\mathbb{T}}$ over the prescribed product. The embeddings remain unchanged at successors and, by finite support, at limits.

For $\lambda \leq \omega_1$, put $\mathbb{C} = \mathbb{C}^n_{E_{<\lambda}}$. In M_n, $\mathbb{C}$ is countably closed and $|\mathbb{C}| \leq \aleph_1$. We may force with $\mathbb{C}$ before Br_n. The countable-closure and finite-product argument from the construction over L gives

$$M_n[K] \models \mathrm{Br}_n \text{ is c.c.c.}, \qquad (\mathrm{Ord}^\omega)^{W_n[K]} = (\mathrm{Ord}^\omega)^{W_n}$$

for generic $K \subseteq \mathbb{C}$, the equality by Easton's lemma. Each almost disjoint factor, including a restriction below a condition, is Knaster. Hence $\mathbb{R}^n_\lambda$ is c.c.c. and has a dense set of finite functions with almost disjoint conditions as values, of size $\leq \aleph_1$. Consequently

$$\operatorname{dens}(\operatorname{Br}_n * \dot{\mathbb{P}}^n_\lambda) \leq \aleph_1, \qquad \operatorname{Br}_n * \dot{\mathbb{P}}^n_\lambda \text{ preserves cardinals and forces } \mathsf{CH}.$$

The same calculation applies to products and to tails extended by trivial factors outside a Boolean value. For an initial-segment quotient, take dense suborders and mixed names for an enumeration of its dense set. This gives the size bound for the combined two-step forcing; thus each such inclusion is a small two-step extension of M_n.

For the localization of reals, take a nice $\mathbb{R}^n_{\omega_1}$-name $\dot{z}$ after forcing with the full Cohen product. Its countable ordinal code lies in W_n by Easton's lemma, and

$$\operatorname{supp}(\dot{z}) = \bigcup\{\operatorname{dom}(p) : (\check{k}, p) \in \dot{z}\} \subseteq \beta \quad \text{for some } \beta < \omega_1.$$

The factors below β use only their own initial segments and $E_{<\beta}$, so the dense suborders of $\mathbb{R}^n_\beta$ agree after $\mathbb{C}^n_{E_{<\beta}}$ and after the full Cohen product. Evaluating the same name, as in Lemma 2.10, therefore gives

$$(2^\omega)^{W_n[G_{\omega_1}]} = \bigcup_{\beta<\omega_1} (2^\omega)^{W_n[G_\beta]}, \qquad G_\beta = G_{\omega_1} \cap \mathbb{P}^n_\beta.$$

Theorem 4.16. *In a generic extension of W_n by a coding iteration of length at most ω_1,*

$$\Phi_n(u) \quad \iff \quad \exists i \ (i \text{ is a coding stage and } u_i = u).$$

Proof. At a coding stage i, the generic real r_i for $\mathbb{A}^n(Y_n(u_i, g_i))$, also below a condition, satisfies $\Psi_n(r_i, u_i)$, hence $\Phi_n(u_i)$, by Lemma 4.15. The selected g_i is obtained from the specified ground-model isomorphism, and the same witness persists to the final extension.

Conversely, suppose $\Psi_n(r, u)$ and no stage codes u. Let K be generic for the full Cohen product, and work in $W_n[K]$. Choose nice names $\dot{r}, \dot{u}$ using the dense suborder of finite functions. Let $D_i \subseteq i$ collect the supports of the names for the parameters of stage i: whether to code and which real, the selected Cohen coordinate and isomorphism, and the condition restricting the almost disjoint factor. Put

$$\begin{aligned} J_0 &= \operatorname{supp}(\dot{r}) \cup \operatorname{supp}(\dot{u}), \\ J_{\ell+1} &= J_\ell \cup \bigcup_{i \in J_\ell} D_i, \qquad\qquad J = \bigcup_{\ell<\omega} J_\ell \in [\omega_1]^{\leq\omega}. \end{aligned}$$

Since $D_i \subseteq J \cap i$ for $i \in J$, induction gives

$$\mathbb{R}^n_{J\cap i} \lessdot \mathbb{R}^n_i, \qquad \dot{r}^{G_J} = r, \quad \dot{u}^{G_J} = u, \qquad G_J = G \cap \mathbb{R}^n_J.$$

At successors the name for the factor uses only earlier stages in J; at limits the embeddings commute with the finite-support projections. The names have a common countable ordinal code z. Easton's lemma and the c.c.c. of the branch product give

$$z \in W_n, \qquad z \in M_n[H_B] \quad \text{for some } B \in [\omega_1]^{\leq\omega} \cap M_n.$$

Let $\mathcal{C} = \{i \in J : i \text{ is a coding stage}\}$ and $I_i = \mathrm{Sel}(h_i, u_i)$ for $i \in \mathcal{C}$, with $I_i = \varnothing$ otherwise. Distinct Cohen coordinates differ on an unbounded set, and the isomorphisms change them only on bounded sets; thus $i \neq j \in \mathcal{C} \implies \sup(h_i \cap h_j) < \omega_1$. Choose $\eta < \omega_1$ such that $B \subseteq \eta$ and $h_i \cap h_j \subseteq \eta$ for all distinct $i, j \in \mathcal{C}$. Lemma 4.15 supplies an unbounded h_r and branches in $L[r, m^n_\infty]$ through every $S^n_{\tau_u(\gamma,k)}$, $\gamma \in h_r$, $k < \omega$. Choose $\gamma \in h_r \setminus (\eta + 1)$; then $|\{i \in \mathcal{C} : \gamma \in h_i\}| \leq 1$. If this set is $\{i\}$, choose k with $u_i(k) \neq u(k)$; otherwise put $k = 0$. Consequently

$$\xi = \tau_u(\gamma, k) \notin B \cup \bigcup_{i \in J} I_i.$$

Put $M_\xi = M_n[K][H \restriction (\omega_1 \setminus \{\xi\})]$; then $z \in M_\xi$. As in Lemma 2.11, interpret the names on J over M_ξ, replacing every factor from the first stage requiring the branch at ξ onward by the trivial forcing. Denote the result by $\widetilde{\mathbb{R}}^\xi_J$. Before that stage, all data defining Y_n lie in the model. Induction on the common dense suborders of finite functions gives, in $W_n[K]$,

$$d_\xi = \bigwedge_{i \in J} \|\xi \notin \dot{I}_i\|_{\mathrm{RO}(\mathbb{R}^n_J)},$$
$$\widetilde{\mathbb{R}}^\xi_J \lessdot \mathbb{R}^n_J, \qquad \widetilde{\mathbb{R}}^\xi_J \restriction d_\xi \simeq \mathbb{R}^n_J \restriction d_\xi.$$

At successors, membership and order in the next factor agree in common generic extensions of the preceding stages. At limits, finite supports are bounded and the corresponding Boolean values are meets. In the given extension, $\xi \notin I_i$ for all $i \in J$; the induced M_ξ-generic $\widetilde{G}$ therefore satisfies $r, u \in M_\xi[\widetilde{G}]$.

The Cohen product can precede the branch forcing, so $M_\xi \models$ "S^n_ξ is Suslin." The Knaster factors and finite support give

$$M_\xi \models \widetilde{\mathbb{R}}^\xi_J \times \mathbb{S}^n_\xi \text{ is c.c.c.} \quad \implies \quad M_\xi[\widetilde{G}] \models \text{"}S^n_\xi \text{ is Suslin."}$$

But $m^n_\infty \in M_\xi$, whence $L[r, m^n_\infty] \subseteq M_\xi[\widetilde{G}]$; the branch through S^n_ξ decoded there contradicts this conclusion. □

4.5 Transfer of the uniformization construction

Fix a recursive universal enumeration of Π^1_{n+3} formulas with two free real variables, and write A^n_m for the corresponding relations. We use the construction in the proof of Theorem 3.1 with W_n, A^n_m, and Φ_n in place of W, A_m, and Φ. We verify the required properties below.

The preceding arguments give the complete embeddings, products, restrictions, and continuations used in Theorem 3.1. The density bound supplies a set of names for tails and choice functions. If N is the common initial extension and $H_{\mathbb{R}}, H_{\mathbb{Q}}$ are the two quotient generics, then $N[H_{\mathbb{R}}] \cap N[H_{\mathbb{Q}}] = N$. Thus the intersection lemma applies to both the choice functions and the restrictions on coding reals outside N. The classes of allowable iterations are defined from the forcing relation for A^n_m and these embeddings.

For persistence write

$$\neg A^n_m(x,y) \iff \exists z\, \theta_m(z,x,y), \qquad \theta_m \in \Pi^1_{n+2}.$$

For the extension $N_{\mathbb{R}} = N[H_{\mathbb{R}}]$ by one factor and the product extension $N_{\mathbb{R},\mathbb{Q}} = N[H_{\mathbb{R}}, H_{\mathbb{Q}}]$, Theorem 4.6 gives

$$\begin{aligned} N_{\mathbb{R}} \models \theta_m(z,x,y) &\Longrightarrow N_{\mathbb{R},\mathbb{Q}} \models \theta_m(z,x,y) \\ &\Longrightarrow N_{\mathbb{R},\mathbb{Q}} \models \neg A^n_m(x,y). \end{aligned}$$

This also applies after appending a tail forcing nonmembership to one factor, including after further coding restrictions. Conversely, if a selected value y satisfies $(x,y) \notin A^n_m$ in the final model, localization and the same absoluteness give

$$\begin{aligned} W_n[G_{\omega_1}] \models \neg A^n_m(x,y) \Longrightarrow \exists \beta < \omega_1\ \exists z \in W_n[G_\beta] \\ W_n[G_\beta] \models \theta_m(z,x,y), \end{aligned}$$

with $x, y \in W_n[G_\beta]$. All these inclusions are small two-step extensions by the size calculation. Hence the arguments for persistence, least attainable ranks, their minimizers, the first segment containing a new minimizer, and preservation of the selected value transfer to M_n.

Theorem 4.17. *Let $1 \le n < \omega$ and suppose that $M^{\#}_n$ exists. There is a cardinal-preserving set-generic extension of M_n satisfying* CH *and the Π^1_{n+3}-uniformization property.*

Proof. Force first with Br_n, then apply Theorem 3.1 over W_n, using the properties verified above. Let $N = W_n[G_{\omega_1}]$ and $\mathcal{C} = \{u_i : i < \omega_1,\ i \text{ is a coding stage}\}$. In N, put

$$U^n_m(x,y) \iff A^n_m(x,y) \wedge \neg\Phi_n(\langle x,y,m\rangle).$$

For $A^n_{m,x} \neq \varnothing$, the construction selects $y_*(m,x) \in A^n_{m,x}$. By Theorem 4.16, for every $y \in A^n_{m,x}$,

$$\neg\Phi_n(\langle x,y,m\rangle) \iff \langle x,y,m\rangle \notin \mathcal{C} \iff y = y_*(m,x).$$

Thus $\{y : U^n_m(x,y)\} = \{y_*(m,x)\}$ for a nonempty section, and equals $\varnothing$ otherwise. Since $A^n_m \in \Pi^1_{n+3}$ and $\Phi_n \in \Sigma^1_{n+3}$ by Lemma 4.15, $U^n_m \in \Pi^1_{n+3}$. Include real parameters in the first coordinate by recursive pairing. Preservation gives $\mathrm{Card}^N = \mathrm{Card}^{M_n}$ and $N \models$ CH. □

5 Further possible applications and open problems

We close with several directions suggested by the construction. First, the method should have an analogue in the generalized Baire-space context at a regular uncountable cardinal κ. In particular, one may ask whether the Π^1_1-uniformization problem for ${}^\kappa\kappa$ can consistently have a positive solution.

The methods in this paper leave the following question open:

Question 1. *Can the Π^1_n-uniformization property be forced over L for $n > 3$?*

The method introduced here is so far limited to local effects. It would be interesting to force a less local, or even global, behaviour:

Question 2. *Given a pair $n, m \in \omega$ such that $n \neq m$ and $n \neq m + 1$, can one force a universe in which the Π^1_n- and the Π^1_m-uniformization properties both hold?*

Finally, it seems worthwhile to combine the present method with other forcing constructions in order to obtain models with properties that are usually consequences of determinacy assumptions. We record one paradigmatic question:

Question 3. *Assume some sufficiently big large cardinal notion which can exist in L. Is there a model which satisfies the Π^1_3-uniformization property and in which every projective set of reals is Lebesgue measurable?*

References

[1] J. Addison. Some consequences of the axiom of constructibility. *Fundamenta Mathematicae*, 46:337–357, 1959.

[2] A. Caicedo and R. Schindler. Projective well-orderings of the reals. *Archive for Mathematical Logic*, 45:783–794, 2006.

[3] R. David. A very absolute Π^1_2-real singleton. *Annals of Mathematical Logic*, 23:101–120, 1982.

[4] V. Fischer and S. D. Friedman. Cardinal characteristics and projective wellorders. *Annals of Pure and Applied Logic*, 161:916–922, 2010.

[5] S. D. Friedman and D. Schrittesser. *Projective Measure without Projective Baire*, volume 1298 of *Memoirs of the American Mathematical Society*. American Mathematical Society, 2020.

[6] Stefan Hoffelner. Forcing the Σ^1_3-separation property. *Journal of Mathematical Logic*, 22(2):2250008, 2022.

[7] Stefan Hoffelner. A universe with large continuum, global Σ-uniformization and a projective well-order of its reals, 2025. Preprint, arXiv:2506.12393.

[8] Stefan Hoffelner and Sandra Müller. On a local variant of the 12th Delfino problem — the Σ-side, 2026. Preprint, arXiv:2606.13199.

[9] T. Jech. *Set Theory*. Springer Monographs in Mathematics. Springer Berlin, 3rd millennium edition, 2003.

[10] R. B. Jensen and R. M. Solovay. Some applications of almost disjoint sets. In *Mathematical Logic and Foundations of Set Theory*, pages 84–104, 1970.

[11] A. Kechris. *Classical Descriptive Set Theory*. Springer, 1995.

[12] Motokiti Kondô. Sur l'uniformisation des complémentaires analytiques et les ensembles projectifs de la seconde classe. *Japanese Journal of Mathematics*, 15:197–230, 1938.

[13] N. Lusin. Sur le problème de M. J. Hadamard d'uniformisation des ensembles. *Comptes Rendus de l'Académie des Sciences*, 190:349–351, 1930.

[14] A. D. R. Mathias. Surrealist landscape with figures. *Periodica Mathematica Hungarica*, 10(2-3):109–175, 1979.

[15] Y. Moschovakis. Uniformization in a playful universe. *Bulletin of the American Mathematical Society*, 77(5):731–736, 1971.

[16] Y. Moschovakis. *Descriptive Set Theory*, volume 155 of *Mathematical Surveys and Monographs*. American Mathematical Society, 2009.

[17] S. Müller, R. Schindler, and W. H. Woodin. Mice with finitely many Woodin cardinals from optimal determinacy hypotheses. *Journal of Mathematical Logic*, 20, 2020.

[18] P. Schlicht. Thin equivalence relations and inner models. *Annals of Pure and Applied Logic*, 165:1577–1625, 2014.

[19] J. Steel. Projectively well-ordered inner models. *Annals of Pure and Applied Logic*, 74:77–104, 1995.

[20] John R. Steel. An outline of inner model theory. In Matthew Foreman and Akihiro Kanamori, editors, *Handbook of Set Theory*, pages 1595–1684. Springer, Dordrecht, 2010. Numbered references use the manuscript dated 26 October 2006.